

Stagflationary Stock Returns^{*}

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Abstract

We study investors’ perceptions of inflation through the lens of a high-frequency event study, documenting they have a stagflationary view of the world. In response to higher-than-expected inflation, investors expect firms’ nominal cash flows to remain stagnant while discount rates increase, lowering stock prices. Both the equity risk premium and nominal risk-free yields rise, but longer-term real yields remain unchanged. Consistent with investors interpreting inflation as a cost shock, investors expect firms with low market power to suffer larger declines in cash flows. Cash flow expectations of equity investors are aligned with those of professional earnings analysts. Our results are at odds with Taylor-rule predictions and support a flat Phillips curve.

JEL Codes: G12, E31, E44, L11

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1 Introduction

The recent inflationary episode has renewed interest in understanding how inflation affects firms and the economy more broadly. Households and firms generally dislike inflation (Shiller, 1997; Stantcheva, 2024) and often hold a stagflationary view of the world, associating inflation with low growth (Candia, Coibion and Gorodnichenko, 2023). How do financial market participants—considered to be more sophisticated agents with “skin in the game”—view the macroeconomic implications of inflation? Do they share the perception of households and firms, or do they take a more demand-side-driven view of inflation? Investors’ perspectives are crucial to understand as they determine asset prices, which in turn affect the cost and allocation of capital in the economy, as well as measures of inflation expectations. Moreover, rich data availability across firms and asset classes allows us to answer these questions using a high-frequency approach around inflationary news and shed light on the channel driving those beliefs.

We show that investors have a stagflationary view of the world. Following inflationary news, investors expect firms’ nominal cash flows to remain stagnant, while inflation expectations and the equity risk premium increase, resulting in lower stock prices. Nominal yields also increase, albeit modestly, driven by inflation compensation rather than real yields. In fact, real yields at policy-sensitive maturities *decline*. Investors therefore seem to reject a Taylor rule response to inflationary news, suggesting they may not consider inflation as demand-driven, which would necessitate a tighter monetary policy stance.

Indeed, one way to rationalize this stagflationary view is that investors interpret inflationary news as supply shocks that increase marginal costs. Canonical models of imperfect competition predict that firms with low market power suffer larger declines in their cash flows in response to inflationary pressures driven by marginal costs. We provide empirical evidence consistent with this prediction of the cost shock hypothesis: Market power shields firms from stagflationary stock returns, with investors expecting high market power firms to generate a relative increase in their cash flows.

Figure I shows that the stock market drops sharply following inflationary news, which is defined as the difference between the inflation print and forecasters’ median expectation. Under the present value formula of the equity price, unexpected stock returns must be due to changes in investors’ expectations of future dividends (cash flow news) or future returns (discount rate news). Nominal interest rates increase with expected inflation (Fisher, 1930), but if stocks’ cash flows are real assets, as conventional wisdom suggests, then nominal expected cash flows should also increase with inflation. The negative stock returns following inflationary news must therefore stem from a combination of declining expectations of future real cash flows and increases in real discount rates, where the latter can occur via increases in investor expectations of future real risk-free yields or because of an increase in the equity risk premium.

To disentangle these three stock return components—cash flow, yield curve, and equity risk premium—we build on Knox and Vissing-Jorgensen (2024), who argue that in modern financial

markets, rich information about the risk-free yield curve and the equity risk premium (via equity option prices) is observable. This allows a decomposition of stock returns into components using the price changes of other financial instruments. The benefit of the observables-based stock decomposition approach is that the discount rate inputs are forward-looking and are also available in real-time at a high frequency. These latter properties make the approach particularly well suited for implementation in an event-study setting such as ours, as compared to traditional vector autoregression (VAR)-based return decomposition methodologies (Campbell, 1991; Campbell and Ammer, 1993).

Before turning to the full return decomposition results, we study how observable discount rates and direct measures of investor cash flow expectations each respond to inflationary news. In their own right, these variables shed light on investors’ perception of the economic implications of inflation. Moreover, they qualitatively inform us about the drivers of stock returns, which we will later quantify.

We start by studying how yields across the term structure respond to inflationary news. We find, unsurprisingly, that nominal interest rates increase in response to inflationary news. However, when decomposing nominal interest rates into real interest rates and inflation compensation, we attribute the entire rise in nominal yields to inflation compensation rather than to an increase in real interest rates. This finding suggests that long-run inflation expectations are not perfectly anchored, but instead are revised in response to inflationary news (Bernanke, 2007). In fact, at policy-sensitive maturities such as the two-year, real yields even *decline* in response to inflationary news. This can be rationalized by investors expecting inflation to be supply-driven and to contract output. In response to a supply-driven increase in inflation, investors may expect the central bank to look through the rise in prices, as it can only control demand. Unchanged long-dated real yields in response to inflationary news also imply that the negative stock returns must come from either (or both) declining real cash flow expectations or an increasing equity risk premium, which immediately reveals a negative interpretation of inflationary news for the marginal equity investor. These results are in stark contrast to monetary policy days, during which real yields instead drive nominal yields (Hanson and Stein, 2015; Nakamura and Steinsson, 2018), which we show, together with higher risk premium, are responsible for the decline in stock prices following monetary policy shocks.

We next study the equity risk premium around inflationary news directly, using the Martin (2017) equity risk premium that is calculated from S&P 500 option prices at the daily level.¹ With this measure, we find evidence that the equity risk premium strongly increases following inflationary news. The effect of inflationary news on the equity risk premium is ambiguous ex-ante, as it depends on the correlation between inflation, economic activity and the asset’s cash flows. If inflation (deflation) is associated with bad states of the world, assets that have low real payouts in inflationary (deflationary) times are relatively undesirable and command higher risk premiums.

¹Knox (2026) provides robust empirical evidence that this option-implied measure of expected returns predicts future realized returns. In section 6 we discuss the underlying theoretical assumptions.

This intuition is embedded in many asset pricing models that explicitly consider the implications of inflation for time-varying risk premiums, such as those with time-varying disaster risk (Gabaix, 2008), long-run risk (Bansal and Shaliastovich, 2013), or risk aversion (Bekaert and Engstrom, 2010; Campbell et al., 2020). Through the lens of these models, an increase in the equity risk premium following inflationary news is consistent with investors interpreting inflation as signaling adverse, stagflationary economic outcomes, rather than stronger real economic activity.

Lastly, we aim to study the role of the expected cash flows for stock returns around inflationary news, but data limitations make the cash flow component the most difficult to study. To address the challenge of obtaining the cash flow component over a long time series and at a daily frequency, we formally decompose stock returns into a yield curve, an equity risk premium, and a cash flow return component. This also allows us to quantify the impact of each return component specifically. We first compute the yield curve component of stock returns, which is defined as the weighted average of changes in interest rates across yield curve maturities, where the weights are the discounted value of dividend payments on the stock market for each corresponding maturity. We next compute the equity risk premium and cash flow return component from a two-stage approach. First, we adjust returns for the observed yield curve return, and, second, we regress the yield-adjusted return component on changes in the observed equity risk premium. The regression-fitted value provides our equity risk premium return component. By the definition of the present value formula, once the discount rate component of stock returns is accounted for, the remainder is attributed to the cash flow return component. This component captures stock returns due to changes in investors' expectations of future cash flows. Those can be nominal or real cash flow expectations, depending on whether we implement the yield curve return component using nominal or real yields.

With three stock return components available, we then regress each return component on inflationary news separately. Following a one percentage point inflation surprise, we observe a 1.42 percent one-day negative nominal return on the stock market, of which we attribute negative 1.05 percentage points to an increasing equity risk premium, 0.61 percentage points to increasing nominal yields, and a small and insignificant positive 0.24 percentage points to increasing nominal cash flow expectations. Real stock prices decline even more. Adjusting for the higher price level, the stock market declines by 2.42 percent in real terms—we attribute 0.75 percentage points of the decline to an increasing equity risk premium, and 1.83 percentage points to declining *real* cash flow expectations. As real yields decline slightly, they even contribute a positive 0.15 percentage points to real stock returns.

So far, we have argued that inflationary news are interpreted as negative supply shocks and presented evidence consistent with this interpretation. At the same time, this implies they are not interpreted as demand or monetary policy shocks. To more rigorously rule out these channels, we apply our methodology to news that more directly reflect each of these shocks. In particular, we use nonfarm payroll (NFP) employment surprises, which are more likely to reflect demand shocks, and FOMC monetary policy surprises and compare resulting asset-price responses with

those to CPI surprises. First, we show that a one-standard-deviation CPI surprise raises the 10-year nominal yield by only about half as much as an equally sized monetary policy or NFP surprise. Second, the drivers of nominal yields differ fundamentally. While increases in nominal yields following monetary policy and NFP surprises are driven primarily by higher real yields, CPI surprises raise nominal yields through higher breakeven inflation. Consequently, higher real yields weigh on stock prices following both monetary policy and NFP surprises. However, positive employment surprises also generate favorable cash flow news² that offset the increase in discount rates, leaving stock prices little changed on net. These results are consistent with economic theory and, while interesting in their own right, provide validation for our methodology. Moreover, we show that the decomposition’s cash flow component in response to inflationary news is strongly correlated with more direct measures of cash flow expectations based on dividend futures and analysts’ earnings expectations.³

Our baseline result that investors interpret inflationary news as stagflationary reflects the average response across a wide range of macroeconomic environments. Our hypothesis is that the Phillips curve is perceived to be sufficiently flat that ordinary fluctuations in demand generate only modest movements in inflation.⁴ However, during episodes when demand is the dominant driver of macroeconomic dynamics, such as the Global Financial Crisis (GFC), even with a flat Phillips curve, inflation can be demand-driven. We provide evidence in favor of those hypotheses. First, we show that investors consider inflation indeed to be demand-driven during the GFC, with the stock market and cash flow expectations positively associated with inflation surprises. By contrast, inflationary news is interpreted more adversely during periods of elevated oil-price volatility, when supply disruptions are especially salient and investors expect larger declines in real cash flows. Second, evidence from the Survey of Professional Forecasters mirrors these patterns: While typically the association between revisions in expected inflation and growth is weak, it strengthens during the Global Financial Crisis. Moreover, forecasters perceive this correlation to be negative when oil-price volatility is high. These findings indicate that investors do not attach a fixed interpretation to inflationary news. Nevertheless, these episodes represent exceptions rather than the rule, and the average stagflationary response remains remarkably robust across other macroeconomic states.

Our results show that both stock and nominal bond returns fall in response to higher-than-expected inflation. This result, at first sight, may appear at odds with the negative stock–bond correlation that prevailed over much of our sample, which has been attributed to inflation being primarily demand driven, see e.g. [Campbell et al. \(2025\)](#) for a review. However, we argue the uncon-

²We view this exercise as a reverse placebo test. A natural concern with our null finding that nominal cash flows do not increase with inflationary news is that the cash flow component contains measurement error and therefore fails to respond even when investors’ true cash flow expectations do. The idea is therefore to test the cash flow component in an alternative setting where the null is expected to be rejected.

³We also use dividend futures ([Gormsen and Koijen, 2020](#)) and analysts’ earnings expectations ([La Porta, 1996](#); [Weber, 2018](#); [De La O and Myers, 2021](#); [Bordalo et al., 2024](#)) as direct measures of cash flow expectations. Both datasets confirm the conclusions from the decomposition, although the relatively short dividend futures sample and the lower frequency of earnings expectations limit the strength of the evidence.

⁴This is consistent with the macro literature findings of a flat Phillips curve, see e.g. [Hazell et al. \(2022\)](#).

ditional stock–bond correlation is not a sufficient statistic to derive the nature of inflation shocks, as it reflects not only inflation news but also other forces affecting bond and stock prices. Indeed, a growing literature finds that changes in macroeconomic fundamentals alone cannot account for the evolution of the stock–bond correlation (Baele et al., 2010; Duffee, 2023; Antolin-Diaz, 2025). Our empirical setting provides a direct test of how investors interpret inflationary news because macroeconomic announcement surprises isolate unexpected inflation innovations. If inflationary news were predominantly interpreted as demand driven, one would expect inflation surprises to raise real yields and expected cash flows.⁵ Indeed, we observe these exact dynamics in asset prices in response to NFP news, but the opposite for inflationary news. Those results suggest that investors interpret inflationary news as negative supply shocks and NFP news as positive demand shocks. Taken together, this evidence suggests that inflationary news itself was not the primary driver of the negative unconditional stock–bond correlation through the majority of our sample. Rather, that correlation is likely shaped by other forces, including time variation in the hedging value of Treasuries (Acharya and Laarits, forthcoming), persistent consumption shocks affecting discount rates (Chernov et al., forthcoming), and changes in monetary policy (Cieslak and Pang, 2021). These channels can generate a negative unconditional stock–bond correlation even when inflationary news is, on average, interpreted as stagflationary.

That said, a caveat to leveraging inflation surprises around CPI announcement days to infer investors’ perceptions of inflation is that investors also receive inflation-related information from other macroeconomic shock events, such as oil price shocks, FOMC meetings, or labor market announcements. This raises the question whether investors consistently interpret inflation as stemming from supply shocks outside CPI announcements, or if they solely consider CPI forecast errors as supply-driven. Addressing this question is challenging due to the endogenous nature of inflation developments, where inflationary news is often intertwined with broader economic updates. Our approach therefore aligns with the literature focusing on monetary policy shocks rather than systematic changes in monetary policy, where disentangling the distinct effects of shocks versus systematic changes poses similar challenges. Focusing on CPI announcement days offers significant advantages, particularly in mitigating endogeneity concerns. By isolating the impact of unanticipated inflation surprises, our event-study framework minimizes biases from the simultaneous occurrence of other macroeconomic shocks.

As discussed, one potential explanation for stagflationary stock returns is that the marginal investor views the Phillips curve as flat. In this world, fluctuations in demand generate only modest movements in inflation, increasing the importance of cost-push disturbances. This interpretation likely has important implications across the firm distribution. Using a standard model of imperfect competition (Tirole, 1989), we show that an increase in marginal costs lowers firm profits, but that the magnitude of the decline depends critically on the elasticity of demand the firm faces, and thus on its market power. Firms with greater market power, and therefore lower demand elasticity,

⁵Whether stock returns are positive, on net, depends on which component dominates (Boyd et al., 2005).

experience smaller declines in profitability in response to higher marginal costs. We test this hypothesis by leveraging cross-sectional variation in stock prices, expected cash flows, and earnings expectations across the firms’ market power distribution. These tests further isolate the impact through cash flow expectations and shed light on the channel behind the negative stock returns around inflationary news. Cross-sectional heterogeneity across firms also allows us to control for various time-varying factors that could be correlated with confounding factors in the time series results.

We measure market power using firm-level markups estimated from a production-function approach following [De Loecker and Warzynski \(2012\)](#) and [De Loecker, Eeckhout and Unger \(2020\)](#).⁶ Consistent with inflationary news representing an adverse marginal cost shock, firms with little market power experience substantially larger stock-price declines than firms with high markups, whose returns are only weakly affected by inflation surprises. This differential response remains after controlling for firm characteristics, factor exposures, and time-varying industry effects, and is not preceded by differential pre-trends. To distinguish between cash-flow and discount-rate channels, we develop a firm-level stock return decomposition, analogous to our aggregate decomposition, that removes the effects of observable discount-rate movements and firms’ heterogeneous sensitivities to those movements. The results show that inflationary news leads investors to revise expected real cash flows downward primarily for firms with low market power, while firms at the top of the markup distribution exhibit little evidence of declining expected real cash flows. We corroborate this interpretation using analyst earnings expectations: although analysts do not revise nominal earnings upward on average following inflationary news, they expect significantly stronger earnings growth for firms with greater market power. Together, these findings indicate that market power insulates firms from the adverse real cash-flow consequences of inflationary news, consistent with firms with greater pricing power being better able to pass higher costs on to customers.

The rest of the paper is organized as follows. In [section 2](#) we discuss the related literature. In [section 3](#) we present the data. In [section 4](#) we lay out the methodology for the stock return decomposition. In [section 5](#) we describe the empirical strategy. In [section 6](#) we present the time-series results. In [section 7](#) we present the cross-sectional results. In [section 8](#) we conclude.

2 Literature

Our paper relates to the literature on how different economic agents perceive inflation. A large literature studies how inflation expectations affect household choices ([D’Acunto et al., 2023](#); [Bachmann et al., 2015](#); [Coibion et al., 2022, 2023, 2024](#)), trading behavior ([Schnorpfel, Weber and Hackethal, 2024](#)), and firm decisions ([Coibion et al., 2018, 2020](#)), documenting that households and firms often associate higher inflation with worse economic outcomes ([Candia et al., 2023](#)).⁷

⁶As an alternative market power measure, we also use the profit-cost elasticity from a network oligopoly demand system based on [Pellegrino \(2025\)](#).

⁷See [Weber et al. \(2022\)](#) for a review.

We instead infer investors’ perceptions of inflationary news from high-frequency asset prices, which provide incentive-compatible measures of expectations and allow us to study both aggregate asset prices and the cross section of firms.

Our findings provide a rational explanation for the well-documented negative relation between inflation and equity prices (Fama and Schwert, 1977; Firth, 1979; Fama, 1981; Sharpe, 2002; Gourio and Ngo, 2020; Katz et al., 2017; Fang et al., 2022; De la O and Myers, 2024). Unlike money illusion explanations (Modigliani and Cohn, 1979; Summers, 1980; Cohen et al., 2005; Campbell and Vuolteenaho, 2004), we show that inflationary news primarily acts as a cost shock that lowers real cash-flow expectations, with firms possessing greater market power experiencing smaller declines.

Our paper contributes to the literature linking the stock–bond correlation to inflation and real activity (Campbell et al., 2017; Boons et al., 2020; Campbell et al., 2020; Cieslak and Pflueger, 2023; Pflueger, 2025; Campbell et al., 2025; Song, 2017; Campbell et al., 2025). Although the unconditional stock–bond correlation has been negative since the late 1990s, we find that inflationary news generates a positive stock–bond return correlation, implying that the unconditional correlation is not, by itself, likely to be informative about whether inflation is perceived as demand or supply driven.

Finally, our paper also contributes to the cross-sectional asset pricing literature on inflation (Bhamra et al., 2023; Weber, 2015; Rubio Cruz et al., 2023) by showing that market power is a key determinant of firms’ exposure to inflationary news.

3 Data

3.1 Inflationary News and Other Macroeconomic Surprises

Our inflation analysis is based on Consumer Price Index (CPI) releases, which are published by the Bureau of Labor Statistics. We focus on month-on-month headline CPI. Releases are usually published in the second week of the month for the CPI values of the previous month. We construct a measure of inflationary news with each inflation release by subtracting inflation expectations for the release from the actual inflation release data:

$$\text{Inflationary News}_t = \pi_t - E_{t'} [\pi_t | \mathcal{I}_{t'}], \quad (1)$$

where π_t is the first release value of the headline month-on-month CPI and $E_{t'} [.] | \mathcal{I}_{t'}$ is the conditional expectation just prior to the release based on available information $\mathcal{I}_{t'}$ at $t' < t$. To measure conditional expectations, we use Bloomberg median forecasts for each inflation release, which are available from 1997, and supplement this with the median from Haver Analytics’ Money Market Services survey, which extends the sample back to 1977. Row 1 of Table A.VIII provides summary statistics of our key regressor Inflationary News_t.

We supplement the CPI surprises with 22 other macroeconomic data release surprises, which are

also provided by Bloomberg, and [Bauer and Swanson \(2023\)](#) monetary policy surprises. The macro surprises are Unemployment Rate, PCE, Real GDP, Capacity Utilization, Consumer Confidence, Durable Goods, ECI, Factory Orders, Average Hourly Earnings, New Home Sales, Housing Starts, Initial Claims, Industrial Production, Michigan Consumer Sentiment, ISM Manufacturing, Nonfarm Payrolls, Retail Sales, PPI, ISM Services, Core PCE Inflation, ADP Employment, JOLTS. For the monetary policy surprises, data is provided by [Acosta et al. \(2025\)](#) and we use both the FOMC statement surprises and the press conference surprises.

3.2 Asset Price Data

We obtain minute-level high-frequency data from the London Stock Exchange Group (LSEG) Tick History database. We use the E-mini S&P 500, 2-year and 10-year Treasuries, and VIX futures contracts, which are all available since 1997. We supplement this with data on a 2-year inflation swap contract, which is available since 2007.

Daily U.S. firm-level stock returns are obtained from Center for Research in Security Prices (CRSP). We follow standard procedures and use ordinary shares traded on the NYSE, AMEX, and NASDAQ exchanges. We also adjust returns for splits, mergers, or other corporate actions and trim at the top and bottom 0.5 percent to mitigate the effects of outliers on our results.

We obtain real Treasury yields from the Federal Reserve website, which provides real yields for the 2-year maturity up through the 20-year maturity that are estimated from Treasury Inflation-Protected Securities (TIPS) yields ([Gürkaynak et al., 2010](#)).⁸ The data also provides nominal interest rates and implied breakeven inflation, which is the difference between nominal yields and real yields for a given maturity, as well as instantaneous forward rates. The sample begins in 1999, two years after the first TIPS was issued by the U.S. Treasury. We supplement the TIPS yields data with the real yields computed from fixed interest rate swaps and inflation swaps for robustness. This data is taken from Bloomberg and begins in July 2004.

For our daily equity risk premium measure, we use the [Martin \(2017\)](#) lower bound of the equity risk premium that is constructed from option prices on the S&P500. We compute the option-implied equity risk premium at various maturities using option price data obtained from OptionMetrics. The sample begins in 1996, but in some analysis we extend the 1-year equity risk premium sample back to 1987 by regressing option-implied equity premium on the VIX and $\ln VIX$ and back-filling with the predicted component.

Finally, we obtain data on dividend futures, which are claims to dividends on the S&P 500 stock market in a particular year, from Bloomberg.

⁸<https://www.federalreserve.gov/data/tips-yield-curve-and-inflation-compensation.htm>

3.3 Market Power

Market power is a firm’s ability to price above marginal cost (Pindyck and Rubinfeld, 2014; Goolsbee et al., 2012). This price-marginal cost gap is the markup (Syverson, 2019).

We estimate markups using the production approach (Hall, 1988, 2018; De Loecker and Warzynski, 2012; De Loecker et al., 2020). Under cost minimization, the markup equals the output elasticity times the inverse of the variable input’s revenue share. From the cost-minimization problem:

$$\theta_{i,t}^\nu = \frac{1}{\lambda_{i,t}} \frac{P_{i,t}^V V_{i,t}}{Q_{i,t}}, \quad (2)$$

where $\theta_{i,t}^\nu$ is the output elasticity of input $V_{i,t}$, λ the marginal cost, $P_{i,t}^V$ the input price, and $Q_{i,t}$ the output. The markup is:

$$\mu_{i,t} = \frac{P_{i,t}}{\lambda_{i,t}}, \quad (3)$$

where $P_{i,t}$ is the output price. Hence:

$$\mu_{i,t} = \theta_{i,t} \frac{(P_{i,t} Q_{i,t})}{(P_{i,t}^V V_{i,t})}. \quad (4)$$

Following De Loecker et al. (2021), we use Compustat North American fundamentals data. We measure variable inputs using COGS ($P_{i,t}^V V_{i,t}$) and revenues using sales $P_{i,t} Q_{i,t}$. Output elasticities are estimated at the two-digit sector level via parametric production function estimation with a variable input bundle and capital. While concerns exist about production approach validity (Bond et al., 2021; De Ridder et al., 2021), it remains the most feasible method for studying markup consequences in asset pricing. We do not attempt to contribute to markup estimation methodology or evaluate economy-wide markup levels. As an alternative, we use the Generalized Hedonic-Linear (GHL) demand system from Pellegrino (2025).⁹ This model correlates closely with De Loecker et al. (2020) but avoids Bond et al. (2021)’s criticisms.

3.4 Other Firm-Level Financial Data

We obtain firm-level financial data from Compustat for controls in the analysis. We use firm size (log of total assets (AT)), the book equity (CEQ) to market equity (PRCC*CSHO/1000) ratio, tangibility (the ratio of tangible assets (PPENT) to total assets) and leverage (the ratio of current debt (DLC) and the long-term debt (DLTT) to total assets).¹⁰

Motivated by the cross-sectional asset pricing literature, we control for firm-level exposures to factor portfolio returns. We use the Fama and French (1993) three-factor portfolios, the Fama and French (2015) five-factor portfolios, and the Carhart (1997) momentum factor. We obtain

⁹We thank Bruno Pellegrino for this suggestion and sharing the data.

¹⁰The market equity is obtained from CRSP variables. We merge year-end values with the Compustat book equity.

these asset pricing factors from Kenneth French’s website. We implement factor controls using a Fama-Macbeth approach. In the first step, we compute rolling 5-year betas of each stock in the sample with respect to the factor portfolios. We then include the estimated rolling betas—lagged one period—as control variables in the main regression specifications.

3.5 Earnings and Macroeconomic Expectations

We also use survey-based measures of expectations from two complementary sources: analyst earnings forecasts from the Thomson Reuters IBES Estimates Database and macroeconomic forecasts from the Survey of Professional Forecasters (SPF).

Our first source is the Thomson Reuters IBES Estimates Database, which contains analyst earnings-per-share forecasts dating back to 1976 (see, among others, [La Porta, 1996](#); [Weber, 2018](#); [Bordalo et al., 2024](#); [De La O and Myers, 2021](#); [De la O and Myers, 2024](#)). Following the literature, we focus on S&P 500 firms, whose earnings are covered by a relatively large number of analysts. We aggregate individual analyst forecasts into firm-level earnings expectations and construct revisions in earnings expectations around CPI announcements. These revisions provide a direct survey-based measure of how professional equity analysts update their expectations of future corporate earnings in response to inflation news. Because the data are available at the firm level, they also allow us to examine whether earnings revisions differ across firms with different degrees of market power. To facilitate comparison with our aggregate asset-pricing results, we additionally construct an asset-weighted aggregate measure of earnings expectation revisions. Details of the construction of the earnings expectations dataset are provided in [Appendix A](#).

Our second source is the Survey of Professional Forecasters (SPF), which provides individual forecasts of inflation and real GDP growth made by professional macroeconomic forecasters. We use the individual-level survey to study how forecasters revise their expectations following macroeconomic news and, in particular, to estimate the perceived Phillips curve embedded in professional forecasts. By relating revisions in expected inflation to revisions in expected real GDP growth, we obtain a direct measure of whether professional forecasters perceive inflationary news as reflecting demand-side or supply-side developments. This survey evidence complements our asset-pricing analysis by providing an independent measure of the beliefs that underlie investors’ interpretation of inflation news. Details of the SPF data and variable construction are provided in [Appendix B](#).

4 Stock Return Decomposition

In this section, we propose a stock return decomposition around inflationary news based on the methodology developed by [Knox and Vissing-Jorgensen \(2024\)](#). We first lay out how, in theory, realized stock returns can be decomposed into a yield curve, equity risk premium and cash flow component. We then describe how to use observable data on interest rates and the equity risk premium to obtain the realized stock return components.

4.1 Background

Under the present value formula of the stock market, which states that the price of the stock market is the discounted sum of all future expected cash flows, an unexpected return on the aggregate stock market must result from changes in expected future real (nominal) cash flows or changes in future real (nominal) required returns (discount rates) or both. Changes in discount rates can then be split into changes in the risk-free rate component of the discount rate and changes in the excess return component of the discount rate—i.e., the equity risk premium component. [Knox and Vissing-Jorgensen \(2024\)](#) formalize this present value intuition with a decomposition of realized stock market capital gains into a yield curve return component, risk premium return component, and a cash flow return component, which reflect changes in investors' expectations of those components.

[Knox and Vissing-Jorgensen \(2024\)](#) show that a one-period capital gain on the stock market can be estimated as follows:

$$\frac{P_{t+1}}{P_t} \approx 1 + \underbrace{\left[\sum_{n=1}^{\infty} w_t^{(n)} G_{t+1}^{D,(n)} - 1 \right]}_{\text{Cash flow factor}_{t+1}} + \underbrace{\left[\sum_{n=1}^{\infty} w_t^{(n)} \frac{1}{G_{t+1}^{YC,(n)}} - 1 \right]}_{\text{Yield curve factor}_{t+1}} + \underbrace{\left[\sum_{n=1}^{\infty} w_t^{(n)} \frac{1}{G_{t+1}^{EP,(n)}} - 1 \right]}_{\text{Equity risk premium factor}_{t+1}}, \quad (5)$$

where P_t is the price of the stock market index at time t , $w_t^{(n)}$ is the price of the expected dividend payment on the aggregate stock market in n period's time as a fraction of the overall price of the stock market index, (i.e. the weight, in present value terms, that a cash flow in period n contributes to the stock market), and $G_{n,t+1}^X$ for $X = \{D, YC, EP\}$ is the cash flow, risk-free rate, and equity risk premium growth factors for $n = 1, 2, \dots$

$$G_{t+1}^{D,(n)} = \frac{E_{t+1}[D_{t+n+1}]}{E_t[D_{t+n}]} \quad G_{t+1}^{YC,(n)} = \frac{(1 + y_{t+1}^{(n)})^n}{(1 + y_t^{(n)})^n} \quad G_{t+1}^{EP,(n)} = \frac{\frac{E_{t+1}(R_{t+1,n+1})}{(1 + y_{t+1}^{(n)})^n}}{\frac{E_t(R_{t,n})}{(1 + y_t^{(n)})^n}} \quad (6)$$

where D_{t+n} is the dividend in n periods' time, $y_t^{(n)}$ is the n -year government bond yield at time t , and $R_{t,n}$ is the cumulative hold to maturity return on the n -year dividend strip from time t to n .¹¹

The decomposition in [Equation 5](#) shows how growth in dividend expectations generates positive stock market returns through $G_{n,t+1}^D > 1$, while growth in risk-free rates and equity risk premium generate negative returns through $\frac{1}{G_{n,t+1}^{YC}} < 1$ or, $\frac{1}{G_{n,t+1}^{EP}} < 1$ respectively. The decomposition also highlights the importance of dividend strip weights, $w_t^{(n)}$, determine the relative importance in growth rates of dividends and discount rates of various n maturities. Intuitively, the more a future expected dividend contributes to the stock market in present value terms, the more changes in expectations of that dividend, or its discount rate, matter for the overall price level of the market.

¹¹At the daily frequency, which is our frequency of interest, the stock market capital gain and the stock market return are approximately equal and therefore we use these words interchangeably through the paper.

A key insight of [Knox and Vissing-Jorgensen \(2024\)](#) is that a lot of information about the aggregate stock market discount rates and, to some extent, cash flow expectations, are available in modern financial markets. Furthermore, dividend strip prices (and thus dividend strip weights) can be calculated from dividend futures prices. This observable data means that one can go a long way to decomposing aggregate stock returns as set out in [Equation 5](#) using the prices of assets in other financial markets. These asset prices are available at a daily frequency and are therefore well suited for implementation in a daily return event study setting. We describe the specific data for the return decomposition used in this paper’s implementation in [subsection 3.2](#) and set out details on the implementation in the subsection below.

4.2 Implementation

4.2.1 Yield Curve Return Component

When estimating the return components of the aggregate stock market return, we start with the yield curve return component and utilize the availability of a rich-term structure of both nominal and real government bond yields out to a 20-year maturity ([Gürkaynak et al., 2010](#)) to compute using the definitions in [Equation 5](#) and [Equation 6](#):

$$Return_t^{k,YC} = \sum_{n=1}^{\infty} w_t^{(n)} \frac{1}{G_{t+k}^{YC,(n)}} - 1. \quad (7)$$

As [Gürkaynak et al. \(2010\)](#) yields are available out to 20 years, we need to make an assumption about the changes in forward yields beyond the maximum observed maturity. We follow [Knox and Vissing-Jorgensen \(2024\)](#) and set $G_{n,t+1}^{YC,(n)} = G_{(20),t+1}^{YC}$ for $n > 20$ which is the same as assuming forward rates don’t move beyond the maximum maturity. Economically, the assumption is that monetary policy 20 years out is unaffected by a CPI release today. To the extent that forward risk-free rates beyond the 20-year maturity move in response to inflationary news, our results will understate the role of the yield curve return component in aggregate returns. However, as we discuss later, we find forward rates beyond the 10-year maturity are unchanged following inflationary news.

4.2.2 Equity Risk Premium Return Component

As with the yield curve component, one can utilize observable discount rate data and the definitions in [Equation 5](#) and [Equation 6](#) to compute the risk premium return component:

$$Return_t^{k,EP} = \sum_{n=1}^{\infty} w_t^{(n)} \frac{1}{G_{t+k}^{EP,(n)}} - 1. \quad (8)$$

Using the [Martin \(2017\)](#) lower bound of the equity risk premium as our measure of observable

risk premium, we do not observe the equity risk premium beyond $n = 2$ years.¹² The observable term structure of the equity premium is therefore less rich as compared to the observable term structure of government bond yields.

As we will see later, equity premium changes following inflationary news are large, and the equity premium component is an important factor in stock returns around inflationary news. To capture potential movements in equity premium beyond the observed maturities, we therefore implement a two-step regression approach to estimate the total risk premium return component of stock returns. First, we compute the k -day aggregate return that is *not* due to the observed yield curve return component on each CPI date,

$$Return_t^{k,Adj} = Return_t^k - Return_t^{k,YC},$$

and, second, estimate the following regression for all $k \in [-5, 10]$ across CPI dates:

$$Return_t^{k,Adj} = \left(\mathbf{G}_{\mathbf{t}+\mathbf{k}}^{\mathbf{EP},(\mathbf{n})} \right)' \Theta^k + \epsilon_t^{k,Adj}, \quad (9)$$

where $\mathbf{G}_{\mathbf{t}+\mathbf{k}}^{\mathbf{EP},(\mathbf{n})}$ is a vector of k -day growth rates in the observed equity risk premium of various n maturities. From these estimations, we then define the risk premium component of the k -day aggregate return as the predicted component of the estimation:

$$Return_t^{k,EP} = \left(\mathbf{G}_{\mathbf{n},\mathbf{t}+\mathbf{k}}^{\mathbf{EP}} \right)' \hat{\Theta}^k. \quad (10)$$

This regression approach assigns all yield curve-adjusted stock returns that are correlated with the equity risk premium at observable maturities to the risk premium return component. To the extent that changes in the unobserved longer-dated equity risk premiums are correlated with changes in observed shorter-dated equity risk premiums, the method still captures movements in stock prices that are due to the unobserved component. This is a useful feature of the approach given that long-dated maturities of the equity risk premiums are unobserved in daily data. Moreover, [Knox and Vissing-Jorgensen \(2024\)](#) show that changes in the wedge between true risk premium and the observed [Martin \(2017\)](#) lower bound of risk premium are positively correlated with changes in the lower bound itself. The regression approach therefore also captures changes in risk premium due to the unobserved changes in the wedge that are correlated with the observed risk premium changes.

¹²Equity risk premium is computed from the prices of S&P 500 equity options, where the maturity of the equity risk premium estimate is the same as the maturity of the option's expiration date. Since 2022, equity options have been available up to five years ahead, but for the majority of our sample equity options have been available up to two years ahead.

4.2.3 Cash Flow Component

The final return component of the return decomposition is the cash flow return component, which is due to changes in investors' expectations of future dividend payments. In our baseline results, we define the cash flow component of the k -day aggregate return as the residual component of the estimation of Equation 9:

$$Return_t^{k,CF} = \epsilon_t^{k,Adj} \quad (11)$$

as this part of the stock returns has removed all variation that is due to observable changes in discount rates.

It should be noted that one potential confounding factor in the estimation of Equation 9 are changes in investors' expectations of cash flows themselves. Or, put differently, our method will attribute stock returns due to changes in investors' cash flow expectations that are correlated with equity risk premium to the equity risk premium return component. Given the cash flow return component and equity risk premium return component are likely positively correlated, this means we are understating the role of the cash flow return component. As we discuss later, given the changes in stock prices and discount rates around inflationary news that we observe in practice, this means the main finding that real cash flows decline would only be stronger if the confounding factor was controlled for.

Instead, the main identifying assumption of the analysis is that the discount rate variables used in Equation 9 capture all discount rate changes that impact aggregate stock returns in our event-window estimations. The assumption is analogous to a variance decomposition of stock returns (Campbell, 1991; Campbell and Ammer, 1993), where the choice of variables included in the VAR determines how the model apportions returns between the discount rate and cash flow news (Campbell, Polk and Vuolteenaho, 2010; Engsted, Pedersen and Tanggaard, 2012). However, the predictor variables typically used in variance decomposition of stock returns are not commonly available on a high-frequency basis. Our approach, by utilizing observed changes in discount rates that are available contemporaneously, therefore allows us to maintain the identification benefits of an event-study estimation while decomposing the drivers of stock returns.

5 Empirical Strategy

Our empirical strategy relies on an event-study approach that examines the financial market variables around the announcement of a CPI release. Studying unexpected realizations of inflation relative to expectations is crucial in causally isolating the effect of inflation. Higher inflation and expectations thereof can be correlated with positive macroeconomic environments, which could bias a time-series regression of financial market outcomes on realized inflation. Instead, while inflationary news can be interpreted through different lenses (e.g., supply or demand), they are unlikely to be correlated with other non-inflationary news.

5.1 Time-Series Analysis

The time-series analysis uses an event-study approach to estimate the following sequence of regressions across CPI dates:

$$y_t^k = \alpha^k + \beta^k \text{Inflationary News}_t + \mathbf{X}_t' \gamma + \epsilon_t^k, \quad (12)$$

where y_t^k is the change in an asset price around inflationary news released at 8:30am. In high-frequency specifications, t is measured in minutes and y_t^k is the change from 1 or 5 minutes before the inflation release to k -minutes after the release. In daily specifications, t is measured in days and y_t^k is the change from the close of business the day before the inflation release to k business days after the inflation release. $\mathbf{X}_t$ is a vector of controls where each element is the sum of surprises over the event window for another particular type of macroeconomic or monetary surprise. The coefficient β^k is the effect of inflationary news on the k -period change in the y asset price controlling for other economic surprises. We estimate Equation 12 for a broad range of asset prices and across multiple k .

5.2 Cross-Sectional Analysis

The estimations in Equation 12 ignore cross-sectional dimensions of returns. To test for heterogeneity across firms with different market power, we estimate:

$$y_{i,t}^k = \alpha^k + \beta_1^k \text{Inflationary News}_t * \text{Markup}_{i,y(t)-1} + \alpha_i^k + \alpha_t^k + \mathbf{X}_{i,t}' \gamma^k + \epsilon_{i,t}^k, \quad (13)$$

where $y_{i,t}^k$ is firm i 's cumulative stock return from the close of business the day before inflationary news to the close of business k days after. We interact inflationary news with one-year lagged markups. The interaction coefficient β_1^k indicates whether high-markup firms respond differentially to inflationary news; a positive coefficient implies outperformance.

The cross-sectional specification allows us to include time fixed effects (α_t^k), controlling for unobserved heterogeneity that could bias β_1^k , and firm fixed effects (α_i^k), controlling for time-invariant firm characteristics. We include controls $\mathbf{X}_{i,t}$ to address identification concerns. Since firm characteristics may correlate with markups and respond differentially to inflationary news, we interact firm characteristics (log assets, tangibility, leverage, book-to-market) with inflationary news. We also control for risk exposure using rolling five-year factor betas via a Fama-Macbeth approach, interacting these betas with inflationary news.¹³

While *firm* $\times$ *time* fixed effects would be collinear with our interaction term, we include *industry* $\times$ *time* fixed effects to control for time-varying, industry-specific factors. This also addresses markup estimation concerns (Doraszelski and Jaumandreu, 2021), as we partial out sector-

¹³In our baseline results we use the Fama and French (1993) three-factor model. The results are also robust to Fama and French (1993) plus Carhart (1997) or Fama and French (2015) five-factor models.

specific output elasticities and compare firms within industries.

5.3 Limitations

Throughout the analysis, we infer investors’ perceptions of inflation through the response of asset prices to inflation surprises. However, an important caveat to leveraging inflation surprises around CPI announcement days to infer investors’ perceptions of inflation is that investors also receive inflation-related information from other macroeconomic shock events, such as oil price shocks, FOMC meetings, or labor market announcements. This raises the question of whether investors consistently interpret inflation as stemming from supply shocks outside CPI announcements, or if they solely consider CPI forecast errors as supply-driven. It is possible that the mix of supply and demand shocks in our inflationary news is different to the “unconditional” mix in the underlying data generating process. However, addressing this question is challenging due to the endogenous nature of inflation developments, where inflationary news is often intertwined with broader economic updates.

Our approach therefore aligns with the literature focusing on monetary policy shocks rather than systematic changes in monetary policy, where disentangling the distinct effects of shocks versus systematic changes poses similar challenges. Focusing on CPI announcement days offers significant advantages, particularly in mitigating endogeneity concerns. By isolating the impact of unanticipated inflationary surprises, our event-study framework minimizes biases from the simultaneous occurrence of other macroeconomic shocks.

6 Time-Series Results

6.1 High-frequency Evidence

Figure I plots the regression coefficient β^k of Equation 12 from $k = -30$ to $k = 120$ minutes. The coefficient for $k = 0$ represents the effect of inflationary news from the minute before the inflation release to the minute of the release, where y_t^k is the log change in the Emini S&P500 futures contract scaled by 100 to give the implied return of the S&P500 in percent. The negative coefficient, represented by the circle at minute=0, of -0.96, shows that in response to a one percentage point inflationary news, stock prices fall by around 1%. The shaded area in blue reflects the 90% confidence interval around the point estimate, which ranges from around -0.71% to -1.22%, indicating strong statistical significance. The figure shows how this immediate impact persists over the following minutes and hours. For example, the coefficient of -1.22 at $k = 30$ indicates that stock prices fall by around 1.2% between the minute before the announcement of the CPI release and 30 minutes after.

The estimates before the inflation release show the log difference in Emini futures contract from the minute before the inflation release to $-k$ minutes before the release. For example, the coefficient

$k = -10$ shows the difference in the log futures value at 8:29am relative to 8:19am. Importantly, in these minutes before the announcement of inflationary news, stock returns do not exhibit a trend, as shown by the statistically insignificant coefficients for $k = -1$ to $k = -30$ that are all close to zero. The absence of a relationship between inflationary news and stock returns the minutes before suggests that what we call inflationary news is indeed news to market participants and is not yet expected by investors.

Panel A of [Table I](#) presents the OLS regression coefficient β^k of [Equation 12](#) where y_t^k is the change in the asset price from 5-minutes before the inflation release to $k = 5$ minutes after in columns 1-4 and to $k = 90$ minutes after the release in column (5). Inflationary news produces economically large and statistically significant movements across asset classes in these narrow high-frequency windows. A one-percent increase in inflationary news is associated with a 1.12 percentage point decline in equity prices, increases of 6.65 and 5.75 basis points in the two- and ten-year nominal Treasury yields, respectively, and a 2.34 percentage point increase in the level of the VIX. The somewhat larger response of the two-year yield indicates that the interest-rate adjustment is concentrated at shorter maturities, consistent with inflation news affecting expectations of near-term monetary policy. The increase in the VIX also represents an increase in risk premium. Inflation compensation also responds strongly: the two-year inflation swap rate rises by 31.0 basis points over the 90-minute window, which suggests an increase in investors' inflation expectations.¹⁴ ¹⁵ All five coefficients are statistically significant at the 1% level using robust standard errors. In fact, the headline inflationary news regressor used in our regression alone explains a substantial share of high-frequency asset-price variation around releases, with R^2 values ranging from 7.6% for the VIX to 32.4% for the two-year inflation swap rate.

As emphasized by [Gurkaynak et al. \(2020\)](#), macroeconomic announcements contain information beyond the headline surprise, and omitting this partially measured news can bias event-study estimates. To address this concern, Panel B of [Table I](#) reports heteroskedasticity-based maximum-likelihood estimates following [Gurkaynak et al. \(2020\)](#), which extract a latent news factor from the elevated variance of asset prices on announcement days relative to non-announcement days.¹⁶ We estimate this factor jointly across a broad cross-section of assets (the S&P 500, the VIX, the two- and ten-year Treasury yields). The single latent factor absorbs a large share of the announcement-day variation in each asset, raising the R^2 substantially relative to Panel A. Crucially, the headline inflationary-news coefficients in Panel B remain nearly identical to their OLS counterparts in Panel

¹⁴The high-frequency inflation swap sample starts in 2007, which is later than the other assets. This asset market is also typically less liquid, which is why we use a wider event window. We drop days from the inflation swap sample with particularly sparse data, which we define as days where there are less than 20 unique swap prices.

¹⁵[Chaudhary and Marrow \(2024\)](#) show that the daily correlation between inflation compensation and stock returns has been positive in recent years, including on CPI release days. Our results demonstrate that this positive comovement does not imply that inflationary news is beneficial for equities. Once changes in inflation expectations are instrumented using inflation surprises, the relationship turns negative, indicating that the positive unconditional correlation is driven by other macroeconomic shocks rather than inflationary news itself ([Table A.IX](#)).

¹⁶In the baseline estimation, for the non-announcement windows we use the same time of the day as CPI events but on 1 and 5 business days before and after the actual releases.

A, while their standard errors shrink. Consistent with the interpretation of [Gurkaynak et al. \(2020\)](#), this confirms that our simple OLS estimates are unbiased and that the responses we document reflect the effect of headline inflation news rather than correlated omitted announcements.

6.2 Daily Evidence

The high-frequency evidence robustly shows that stock prices decline, while nominal yields, inflation expectations, and measures of risk increase with inflationary news. Motivated by this evidence, we now turn to daily asset price changes. The benefit of studying daily asset prices is that it expands the set of testable assets, which allows richer analysis for teasing out the economic drivers of asset price changes. In particular, it allows us to decompose the drivers of nominal yields across the term structure, as well decomposing stock returns using the methodology decomposition developed in [section 4](#).¹⁷ This stock return decomposition crucially allows us to estimate changes in investors expectations of cash flows around inflationary news. Furthermore, daily data also allows us to turn to the cross section and study the variation in stock returns across firms in response to inflationary news.

Panel A of [Table II](#) shows that the high-frequency results hold at the daily frequency and in a substantially longer sample extending back to 1977. While the longer sample increases statistical power, this benefit is offset by the wider measurement window that introduces additional variation in asset prices that are unrelated to the inflationary news. Consistent with this deterioration in signal-to-noise, the regressions generally have lower explanatory power and less precise estimates than their high-frequency counterparts in [Table I](#). Nevertheless, the estimated responses remain economically large, and most remain statistically significant, with the magnitudes broadly consistent with the high-frequency evidence. A one percentage point inflationary news is associated with a 0.92 percent decline in equity prices and daily increases of 7.68 and 8.21 basis points in two- and ten-year nominal yields, respectively, and a 23.13 basis point increase in the one-year [Martin \(2017\)](#) option-implied equity risk premium, in the specifications with controls. Because the daily event window may contain additional macroeconomic releases and monetary-policy news, we include controls for these announcements to isolate the effect of inflationary news. Controlling for other macroeconomic and monetary-policy surprises occurring within the event window raises the R^2 meaningfully from 0.021 to 0.073 for equities, from 0.028 to 0.069 for two-year yields, from 0.028 to 0.070 for ten-year yields, and from 0.017 to 0.121 for the one-year equity premium. The inclusion of these controls substantially raises the R^2 while leaving the estimated coefficients essentially unchanged, indicating

¹⁷Unlike the high-frequency analysis, which relies on VIX futures changes for a proxy for changes in risk premia, the daily data allow us to directly observe changes in the [Martin \(2017\)](#) equity risk premium across multiple maturities. This option-implied measure is the perceived equity risk premium of log investor fully invested in the market over that horizon, and is a lower bound of the equity premium under the much weaker assumption that the stock market return is negatively correlated with the stochastic discount factor. [Knox and Vissing-Jorgensen \(2024\)](#) also provide theoretical evidence that *changes* in the observed equity risk premium will likely be a lower bound on changes in the true equity risk premium. Thus, to the extent the lower bound is not exact, our approach will understate the already-large risk premium increases observed with inflationary news.

that the estimates are stable to the inclusion of observables and are therefore unlikely to be driven by omitted contemporaneous macroeconomic announcements.¹⁸

Panel B of [Table II](#) examines asset-price changes over the five trading days beginning on the release date. For equities, the response becomes larger in absolute value: in the specification with controls, the coefficient increases in magnitude from -0.92 over one day to -1.43 over five days. The estimated increase in the one-year equity premium also becomes substantially larger, rising from 23 to 59 basis points. These patterns indicate that the adverse implications of inflation news for equities are not fully incorporated on the release day. They are consistent with evidence of continued adjustment to macroeconomic announcements after the initial release documented by [Brooks et al. \(2018\)](#), [Neuhierl and Weber \(2018\)](#), [Altavilla et al. \(2019\)](#), [Rodnyansky et al. \(2026\)](#), and [Bauer et al. \(2023\)](#). The yield responses follow a different pattern. The two-year nominal-yield response largely dissipates over the five-day window, while the ten-year response remains similar in magnitude but is estimated much less precisely. Thus, most nominal-yield adjustment occurs immediately, whereas the equity-price decline and increase in the equity premium continue to develop over subsequent days.

6.2.1 Interpreting the Equity Risk Premium

Ex ante, the effect of inflationary news on the equity risk premium is ambiguous, as it depends on the relationship between inflation and economic activity, firms’ cash flows, and investors’ risk aversion. A broad class of asset-pricing models predicts that the equity risk premium should rise if inflationary news signals a deterioration in future economic conditions. If inflation is associated with adverse states of the world, assets with low real cash flows during periods of high inflation become less desirable and therefore command higher expected returns. This intuition underlies models with time-varying disaster risk ([Gabaix, 2008](#)), long-run risks ([Bansal and Shaliastovich, 2013](#)), and time-varying risk aversion ([Bekaert and Engstrom, 2010](#); [Campbell et al., 2020](#)), each of which explicitly considers the relation between inflation and risk premiums in equilibrium. Through the lens of these models, an increase in the equity risk premium following inflationary news is evidence that investors associate inflation with adverse macroeconomic outcomes and weaker future real cash flows—that is, a stagflationary view of the world.

The higher equity risk premium may reflect increases in investors’ risk aversion, or their perception of risk in the economy broadly or in stock returns more specifically. To help distinguish among these channels, we employ the decomposition of [Bekaert et al. \(2022\)](#), which separately identifies changes in risk aversion and macroeconomic uncertainty from asset prices. [Table A.XI](#) shows that inflationary news increases both components.

The increase in risk aversion is consistent with market participants decrease willingness to absorb risk in inflationary states, while the increase in countercyclical (“bad”) uncertainty reflects

¹⁸These results are robust to using core inflation surprises instead of headline, as shown in [Table A.X](#). The increase in the R^2 from daily to intraday frequency is consistent with [Gorodnichenko and Weber \(2016\)](#).

their perception of greater downside uncertainty surrounding future real cash flows, consistent with [D’Acunto and Weber \(2024\)](#). Both channels are consistent with investors interpreting inflationary news as stagflationary under the interpretation of the models described above. The increase in macroeconomic uncertainty also appears to be reflected in greater uncertainty about the future conduct of monetary policy. Specifically, inflationary news raises the Kansas City Fed’s Measure of Policy Rate Uncertainty (KC PRU), indicating greater uncertainty about future economic conditions and uncertainty about how policymakers will respond. Importantly, this result does not imply that inflationary news mechanically shifts expected interest rates upward. Instead, it might reflect greater uncertainty about what policymakers consider the appropriate policy response. When inflationary pressures coincide with stronger expected real economic outcomes, the response of monetary policy is more certain due to the so-called “divine coincidence” as the appropriate response on both sides of the central bank’s mandate are the same. On the other hand, when inflationary pressures coincide with weaker expected real economic outcomes, the central bank faces a dilemma. Increasing monetary policy uncertainty with inflationary news is thus further suggestive evidence of a supply shock interpretation.

6.3 Decomposing Nominal Yields

The evidence above showed that nominal Treasury yields increase in response to inflationary news, which is unsurprising given the expected monetary policy response to inflation. Economically, however, the nominal-yield response is difficult to interpret without determining whether it reflects higher inflation compensation, higher real interest rates, or both. [Table III](#) therefore decomposes the one-day change in nominal Treasury yields into changes in breakeven inflation and real yields. The estimation period is 1999–2022, reflecting the availability of inflation-linked Treasury yield data from [Gürkaynak et al. \(2010\)](#). Panel A reports results at the two-year maturity, while Panel B reports results at the 10-year maturity. Within each panel, the odd-numbered columns exclude controls and the even-numbered columns include controls. Because the estimates are similar across specifications, we focus on the specifications with controls.¹⁹

The two maturities provide complementary information. The two-year yield is particularly sensitive to the near-term inflation outlook and the expected path of monetary policy. The 10-year yield is more directly relevant for valuing equities because the aggregate stock market is a long-duration claim ([van Binsbergen, 2020](#); [Knox and Vissing-Jorgensen, 2024](#)). Indeed, [Knox and](#)

¹⁹[Table A.XII](#) presents an analogous decomposition using interest-rate and inflation swaps. This exercise addresses the concern that changes in the liquidity premium on Treasury inflation-protected securities around CPI announcements could drive the bond-based results, given that TIPS are less liquid than nominal Treasuries ([Fleckenstein et al., 2014](#)). The swap-based estimates are similar, providing no evidence that variation in TIPS liquidity accounts for the results. Illiquidity may be particularly important during the early development of the inflation-linked bond market from 1999 to 2003 and during the Global Financial Crisis. In unreported results, the findings are similar when we exclude these periods. The negative response of short-term real yields to inflationary news is also robust across subsamples of the 1999–2022 period. Finally, in [subsection 6.5](#), we show that movements in nominal yields are generally driven by changes in real yields, ruling out daily TIPS illiquidity as an explanation for the muted response of real yields to inflationary news.

Vissing-Jorgensen (2024) show that, in recent years, more than 80 percent of the value of the aggregate stock market reflects the present value of cash flows paid more than 10 years in the future.

Panel A shows that a one-percentage-point increase in inflationary news raises the two-year nominal Treasury yield by 9 basis points. This relatively modest increase in the nominal yield masks a much larger change in its components. Two-year inflation compensation increases by 38 basis points, while the two-year real yield declines by 29 basis points. Thus, the short-maturity nominal-yield response is driven by a pronounced upward revision in inflation compensation that more than offsets a decline in the real yield.²⁰

The Taylor principle states that the nominal interest rate should be raised more than point for point when inflation rises so that the real interest rate increases (Taylor, 1993, 1999). By studying policy-sensitive two-year yields, our analysis is allowing us to estimate investor perceptions of the Taylor rule in response to changes in inflation expectations over time.²¹ Under a Taylor rule, the response of monetary policy—and therefore short-dated nominal interest rates—should exceed the change in inflation expectations. However, in the data, we find that real yields at policy-sensitive maturities decline, as the increase in inflation compensation is larger than that of nominal Treasury yields, which is inconsistent with the Taylor rule hypothesis. This finding can be rationalized by investors expecting inflation to be supply driven, contracting output without requiring a strong increase in the nominal monetary policy rate.

Panel B shows a similar pattern at the 10-year maturity to that observed at the 2-year maturity, although with less pronounced movements. A one-percentage-point increase in Inflationary News raises the 10-year nominal Treasury yield by 6 basis points and 10-year inflation compensation by 8 basis points. This response suggests that long-run inflation expectations are not perfectly anchored,²² but instead are revised in response to inflationary news (Gürkaynak et al., 2005; Carvalho et al., 2023; Coibion and Gorodnichenko, 2025). The estimated change in the 10-year real yield is thus small and statistically indistinguishable from zero and, if anything, decreases. The increase in the long-term nominal yield therefore reflects an increase in inflation compensation rather than an increase in the long-term real interest rate.

The nominal yield decomposition is important for interpreting the negative stock-market response to inflationary news. Although nominal yields increase, the evidence does not indicate that inflationary news raises the long-term real discount rate applied to equity cash flows. Long-term real yields are essentially unchanged, while short-term real yields decline. Consequently, the de-

²⁰The strong response of inflation compensation is consistent with prior evidence that inflation news affects both market-based inflation compensation and the inflation expectations of households and professional forecasters (Bauer, 2015; Skaperdas, 2023). In unreported results, we confirm that Blue Chip inflation expectations rise in response to inflationary news. Similarly, D’Acunto et al. (2023) show that when households’ short run inflation expectations increase, they typically also increase their medium and longer term inflation expectations.

²¹The exercise is therefore similar in spirit to recent work that studies professional forecasters’ perceptions of the Taylor rule from surveys (Bauer et al., 2022).

²²Bernanke (2007) defines inflation expectations as “anchored” when they are “relatively insensitive to incoming data.”

cline in stock prices following inflationary news cannot be attributed to an increase in Treasury real rates; if anything, the real-yield response works against the observed decline in equity values. We expand on the implications of these findings in the next section.

6.4 Cash Flow Expectations

To study investors' cash flow expectations, we implement the stock return decomposition methodology described in [section 4](#). This method allows us to quantify the impact of changing discount rates on the overall stock market return, and extract a residual, which, by definition, is due to changes in investors' expectations of future cash flows. Whether these are real or nominal cash flow expectations depends on whether we quantify the impact of real risk-free rates or nominal risk-free rates when calculating the yield curve return component of the aggregate stock return. Either way, realized returns of the stock market are the sum of the yield curve return component, the equity risk premium return component, and the cash flow return component. This means that the beta of the stock market return on inflationary news is by definition equal to the sum of the beta of the return components on inflationary news. The return decomposition methodology therefore allows us to quantify the relative importance of the various return components to aggregate stock market returns in response to inflationary news.

6.4.1 Nominal Stock Return Decomposition

Columns (1)–(4) of [Table IV](#) decompose the one-day response of nominal stock returns to inflationary news into nominal yield curve, equity risk premium, and nominal cash flow components. Because the return components are generated from the first-stage decomposition, the reported standard errors are based on 2,000 bootstrap replications. In each replication, we re-estimate the return decomposition, regenerate the equity risk premium and cash flow components, and then re-estimate their responses to inflationary news.

A one-percentage-point positive inflation surprise lowers nominal stock returns by 1.42 percentage points on the announcement day.²³ The nominal yield curve component contributes -0.61 percentage points, consistent with the increase in nominal Treasury yields documented in [Table III](#). Higher nominal risk-free rates increase the rate at which future cash flows are discounted and therefore reduce current stock prices. Quantitatively, however, this contribution is relatively modest and statistically insignificant.²⁴

²³The aggregate stock return response is more negative than in [Table II](#) because the decomposition sample begins in 1999, when the required discount-rate data become available. Stock returns respond more negatively to inflationary news in this post-1999 sample.

²⁴As discussed in [section 4](#), our methodology could understate the role of yield curve news if forward nominal yields beyond the maximum observed maturity of 20 years respond to inflationary news. The evidence in [Figure A2](#) does not support this possibility: the response of observed forward yields is concentrated at shorter horizons and is negligible beyond the 10-year horizon. [Nagel and Xu \(2024\)](#) document similar maturity patterns in the response of forward rates to monetary policy news.

The equity risk premium component is economically larger, contributing -1.05 percentage points, although the estimate is imprecisely estimated using the bootstrap standard error. By contrast, the nominal cash flow component is 0.24 percentage points and statistically indistinguishable from zero. Thus, the decline in nominal stock prices reflects higher discount rates, particularly a higher equity risk premium, without an offsetting upward revision in expected nominal cash flows.

The absence of an increase in expected nominal cash flows is important for interpreting the stock market response. If firms could pass the increase in the aggregate price level fully into their nominal revenues and profits, expected nominal cash flows would rise with the price level. Instead, the estimates indicate that nominal cash flow expectations remain broadly unchanged. Consequently, the increase in the price level reduces the expected purchasing power of those cash flows, a prediction that we examine directly using the real return decomposition.

6.4.2 Real Stock Return Decomposition

We next decompose real stock returns to extract the response of investors' expectations of future real cash flows to inflationary news. We construct the real stock return by subtracting the inflation surprise from the nominal stock return. This adjustment removes the mechanical effect that a one-for-one increase in expected nominal cash flows with the price level would have on nominal stock values. In particular, if equities were perfect real assets whose nominal cash flows increased proportionally with the aggregate price level, this adjustment would leave their real value unchanged.

Columns (5)–(8) of [Table IV](#) present the resulting decomposition. A one-percentage-point positive inflation surprise lowers the real stock return by 2.42 percentage points on the announcement day. The real yield curve component is positive but small, contributing 0.15 percentage points, and is statistically insignificant. Real rates therefore slightly cushion the fall in stock prices rather than causing it. The equity risk premium component contributes -0.75 percentage points, although this estimate is also statistically insignificant.

Most importantly, the real cash flow component falls by 1.83 percentage points and is statistically significant at the 1 percent level using the bootstrap standard error. The point estimate accounts for approximately three quarters of the decline in the real stock return. The one-day decline in real stock prices therefore primarily reflects a downward revision in investors' expectations of firms' future real cash flows, accompanied by an economically meaningful, though imprecisely estimated, increase in the equity risk premium.

The contrast between the nominal and real cash flow decompositions is central to our interpretation. Expected nominal cash flows do not respond significantly to inflationary news, whereas expected real cash flows decline sharply. Investors therefore do not expect firms' nominal cash flows to increase sufficiently to compensate for the higher price level. This evidence is inconsistent with equities providing a short-run hedge against unexpected inflation and is instead consistent with inflationary news being interpreted as adverse news about firms' real profitability.

To illustrate the dynamics underlying this result, [Figure II](#) presents the real return decom-

position from five days before through ten days after the announcement. The figure shows little systematic movement before the CPI announcement. Following the announcement, the real cash flow component declines immediately and remains persistently negative, while the equity risk premium component becomes increasingly negative over the subsequent days. The real yield curve component remains positive and comparatively small. Thus, the persistent decline in real stock prices reflects lower expected real cash flows and a higher equity risk premium, rather than higher real risk-free rates.

6.4.3 Validation of Cash Flow Component

Assigning the residual of the stock return decomposition to the cash flow component may raise the concern that it does not actually measure news about cash flows but rather reflects unobserved or mismeasured discount rate movements. We therefore provide several validation exercises to verify our decomposition approach.

First, we compare our cash flow component to analysts' earnings expectations aggregated over 2-week windows around CPI releases. [Figure A3](#) (left panels) shows strong correlations between both same-day and 5-day changes in our estimated cash flow component and changes in analyst earnings forecasts. Second, we compare our cash flow component to dividend futures price changes adjusted for the [Martin \(2017\)](#) equity risk premium following [Ibert et al. \(2022\)](#) and [Knox and Vissing-Jorgensen \(2024\)](#).²⁵ [Figure A3](#) (middle and right panels) shows strong correlations in both nominal and real terms, despite data limitations (available only from 2016 with limited liquidity). Third, in unreported results, we implement a [Campbell and Ammer \(1993\)](#) VAR decomposition following [Bernanke and Kuttner \(2005\)](#).²⁶ This corroborates the negative cash-flow response but suffers from low power at monthly frequency and conditional interpretation concerns discussed by [Nagel and Xu \(2024\)](#).

These validations confirm our estimated cash flow component captures market participants' views of the cash flow outlook, and seems aligned with those of professional analysts. In the next section, we study how the decomposition behaves around different macroeconomic announcements, which both provide independent insights and serve as an additional sanity check for our methodology.

6.5 Decomposition Around Different Macro Surprises

So far, we have argued that inflationary news are interpreted as negative supply shocks and presented evidence consistent with this interpretation. At the same time, this implies inflationary news are not interpreted as demand or monetary policy shocks. To more rigorously rule out these

²⁵[Appendix C](#) provides further analysis of dividend futures price changes in response to inflationary news, with the finding supporting the conclusion that investors' nominal (real) cash flow expectations are stagnant (declining) when inflation is higher-than-expected.

²⁶See [Appendix D](#).

channels, we apply our methodology to news that more directly reflect each of these shocks. In particular, we use nonfarm payroll (NFP) employment surprises, which are more likely to reflect demand shocks, and FOMC monetary policy surprises and compare resulting asset-price responses with those to CPI surprises.

Figure III presents the decomposition results for all three announcement types. The left panel decomposes changes in the 10-year nominal Treasury yield into real yields and inflation compensation, while the right panel decomposes real stock returns into their underlying components. FOMC surprises are measured using the high-frequency monetary policy surprise of [Bauer and Swanson \(2023\)](#), where a positive surprise corresponds to an unexpected monetary policy tightening. NFP and CPI surprises are measured relative to the median forecaster expectation.²⁷ For ease of comparison, we standardize each surprise series to have unit variance.

The yield decomposition reveals several important differences across announcement types. Positive surprises raise 10-year nominal yields in all three cases, but the underlying drivers differ substantially. Following an unexpected monetary policy tightening, real yields increase by more than nominal yields, consistent with tighter monetary policy, as inflation expectations fall.²⁸ In contrast, positive NFP surprises raise nominal yields by more than real yields, consistent with stronger economic activity being accompanied by higher inflation expectations. That said, the increase in nominal yields is still primarily driven by higher real yields, with inflation expectations moving relatively little, consistent with evidence that variation in nominal Treasury yields is generally not dominated by movements in inflation expectations ([Duffee, 2018](#)). The response to inflationary news is notably different. First, the increase in nominal yields is relatively muted compared with monetary policy and employment surprises. A one-standard-deviation CPI surprise raises the 10-year nominal yield by only about half as much as an equally sized monetary policy or NFP surprise. Second, unlike the other announcements, the rise in nominal yields is driven almost entirely by inflation compensation, with little contribution from real yields, suggesting that long-run inflation expectations are not perfectly anchored, in the sense of [Bernanke \(2007\)](#), as they respond to inflationary news ([Gürkaynak et al., 2005](#); [Carvalho et al., 2023](#); [Coibion and Gorodnichenko, 2025](#)). These differences provide further evidence that CPI releases convey information that is distinct from both monetary policy and other macroeconomic news.

The stock return decomposition mirrors these differences. Consistent with the rise in real yields, the yield curve component contributes negatively to equity returns following both monetary policy and NFP surprises. However, the overall stock market response differs markedly across the two announcements. Following an unexpected monetary policy tightening, equity prices decline, and

²⁷We drop the May, June, and July 2020 observations from the NFP sample because the corresponding surprises were extreme outliers of 13, 89, and 14 standard deviations, respectively.

²⁸This finding is consistent with [Hanson and Stein \(2015\)](#) and [Nakamura and Steinsson \(2018\)](#), who show that monetary policy surprises move long-dated nominal yields almost exclusively through real yields. In [Appendix E](#), we reconcile our results with this literature by replicating the [Hanson and Stein \(2015\)](#) finding that, on FOMC announcement days, long-dated nominal and real forward yields move in lockstep. Repeating the same exercise on CPI announcement days, however, shows that the response of long-dated nominal forward rates is instead driven primarily by inflation compensation.

this decline is almost entirely explained by higher discount rates both through the yield curve component, closely matching the findings of [Nagel and Xu \(2024\)](#) and [Knox and Vissing-Jorgensen \(2025\)](#), and an increase in the risk premium, consistent with [Bauer et al. \(2023\)](#).

The response to employment surprises is fundamentally different. Positive employment surprises increase equity prices slightly despite the negative effect of higher real yields. Our decomposition shows that expected cash flows rise substantially more than the raw stock return suggests because part of the positive cash flow news is offset by the endogenous increase in discount rates. This finding is consistent with [Boyd et al. \(2005\)](#) and illustrates an important advantage of our methodology. By separating discount rate effects from cash flow revisions, the decomposition uncovers the positive news about future corporate cash flows implied by stronger-than-expected economic activity.

These alternative decompositions provide a useful benchmark for interpreting the response to inflationary news. Monetary policy surprises primarily affect equity prices through higher real discount rates, while positive employment surprises generate favorable cash flow news that is partially offset by higher real discount rates. Inflationary news, by contrast, is characterized by a sharp deterioration in expected real cash flows. This distinct pattern reinforces our conclusion that CPI announcements convey information beyond conventional monetary policy or growth news.

The fact that positive employment surprises generate favorable cash flow news in the decomposition can be viewed as a “reverse placebo” test. Rather than expecting a null result, as in a conventional placebo test, we expect to find a positive cash flow response to news that is naturally interpreted as favorable for future economic activity. In [Figure A4](#), we show that this result is robust across horizons and is corroborated by professional analysts, who also revise upward their expectations of future nominal cash flows and corporate earnings following positive employment surprises. In other words, when the economy creates more jobs than expected, both market prices and analyst forecasts incorporate stronger future cash flows. By contrast, neither investors nor analysts revise cash flow expectations upward following positive inflation surprises. The absence of a similar response following inflationary news therefore strengthens our interpretation that CPI announcements are associated with lower expected real cash flows rather than reflecting a failure of the decomposition.

6.6 State-Dependence

The results thus far describe that the *average* market response to inflationary news is stagflationary. One interpretation is that the Phillips curve is sufficiently flat that ordinary fluctuations in demand generate only modest movements in inflation, increasing the relative importance of non-demand forces, including cost-push disturbances ([Hazell et al., 2022](#)). Indeed, inflation has been shown to be on average supply driven in estimated New Keynesian models that exhibit relatively flat Phillips curves ([Smets and Wouters, 2007](#); [Fratto and Uhlig, 2020](#)). Nevertheless, as long as the Phillips curve is not completely flat, sufficiently large demand shocks should still impact inflation, and therefore periods dominated by large demand shocks can still lead to demand-driven inflation. The

Global Financial Crisis (GFC) is a prime example in our sample where one might expect this to be the case. Periods of elevated oil-price volatility, by contrast, make input-cost and supply conditions salient, raising the likelihood that an inflation surprise is perceived as particularly adverse supply-side news.

To formally test these two predictions, we augment the baseline decomposition with interaction terms that allow the response to inflation news to vary across the GFC recession and periods of elevated oil-price volatility. The point estimates in [Table V](#) support both predictions. During the GFC, the nominal stock-return response to inflation news turns substantially more positive: the interaction with the GFC indicator is more than offsetting the negative baseline, with both the equity-premium and cash-flow interactions positive. When oil-price volatility is elevated, the response is markedly more negative, reflecting a stronger contribution from negative equity premium and cash flow components. These results indicate that investors do not attach a fixed interpretation to all inflation surprises. Inflationary news was viewed more favorably during the GFC, when deficient demand was the principal macroeconomic concern, and more adversely when volatile energy prices make cost and supply disturbances especially salient.

We find complementary evidence in the expectations of professional forecasters. Using individual-level Survey of Professional Forecasters data, we estimate the comovement between revisions in four-quarter-ahead expected CPI inflation and revisions in expected real GDP growth, including forecaster and quarter fixed effects. The unconditional relation is essentially zero ([Table A.XIII](#), column 1). During the GFC, however, the slope increases, indicating that forecasters who revise expected growth upward also revise expected inflation upward. By contrast, oil-price volatility reduces the inflation–growth slope. The survey evidence therefore echoes the state dependence observed in asset prices: inflation is more closely associated with stronger expected growth when deficient demand is especially salient, whereas this positive comovement weakens when energy-price volatility is high.²⁹

Taken together, the asset-price and survey evidence indicate that the interpretation of inflationary news is state dependent. Nevertheless, the average response to CPI surprises remains stagflationary: higher-than-expected inflation is associated with lower expected real cash flows and higher required returns. [Appendix F](#) shows that this negative average response is robust across a broader range of periods and state indicators.

6.7 Discussion of Stock-bond Correlation

At first sight, our finding that investors perceive inflation as supply-driven, on average, may appear at odds with the negative stock-bond correlation observed between the late 1990s and 2020, which has previously been attributed with demand-driven inflation (see [Campbell et al. \(2025\)](#) for a

²⁹Relatedly, [Bansal and Shaliastovich \(2013\)](#) document a negative association between expected inflation and expected future growth, consistent with the broader stagflationary interpretation of inflation expectations.

review). However, we argue that the stock-bond correlation is not a sufficient statistic for identifying the nature of inflation because many forces beyond inflation news affect bond and stock prices. Consistent with this view, [Baele et al. \(2010\)](#), [Duffee \(2023\)](#) and [Antolin-Diaz \(2025\)](#) find that macroeconomic fundamentals explain only part of the substantial time variation in stock-bond correlations.

In fact, our setting could be viewed as a natural laboratory to test the hypothesis that inflation is perceived to be demand driven, as macroeconomic surprises have been shown to be a vital identification tool for understanding macroeconomic dynamics ([Nakamura and Steinsson, 2018](#); [Bauer and Swanson, 2023](#); [Boehm and Kroner, 2023](#)). If inflation were demand-driven, one would expect real yields and cash flow expectations to rise in response to inflationary news. Instead, we find the opposite. Taken at face value, this evidence suggests that inflationary news is unlikely to be the primary force behind the negative unconditional stock-bond correlation, pointing instead toward alternative channels.

What does drive the stock-bond correlation? Recent work identifies several mechanisms that operate independently of the interpretation of inflation. First, [Acharya and Laarits \(forthcoming\)](#) and [Antolin-Diaz \(2025\)](#) show that during periods of financial stress, Treasuries’ hedging value and convenience yield rise as investors shift away from risky assets, lowering stock prices while raising bond prices.³⁰ Second, [Chernov et al. \(forthcoming\)](#) attribute the post-2000 sign switch to a shift from transitory to persistent consumption shocks, which turns bonds into hedges as weak growth news lowers real rates and raises bond prices; this operates through discount rates, not inflation, and holds for indexed as well as nominal bonds. Third, [Boyd et al. \(2005\)](#) and [Cieslak and Pang \(2021\)](#) show that monetary policy is a key driver of joint stock-bond movements, so shifts in the policy regime can alter their comovement even holding the inflation process fixed.

Our empirical design is not intended to determine which of the above channels explains the largest share of the time variation in the aggregate stock-bond correlation. It establishes the narrower but important result that the negative stock-bond correlation does not by itself imply that inflationary news was perceived as demand-driven. When inflationary news is isolated, the market response is instead stagflationary.³¹

7 Cross-Sectional Results

As discussed, one potential explanation for stagflationary stock returns is that the marginal investor views the Phillips curve as relatively flat. In this world, fluctuations in demand generate only modest movements in inflation, increasing the importance of cost-push disturbances. As a result, inflation surprises are interpreted primarily as shocks to marginal costs. This interpretation has clear cross-

³⁰ [Antolin-Diaz \(2025\)](#) links the increased importance of this channel after the late 1990s to the growing role of levered financial intermediaries and shocks to their risk-bearing capacity.

³¹ We discuss the limitations of this interpretation and of our announcement-based identification in [subsection 5.3](#).

sectional implications: the effect of inflationary surprises should vary systematically with firms' ability to absorb or pass through higher costs.

7.1 Theoretical Framework

To formalize these implications, we consider a canonical model of imperfect competition in which firms set prices as the product of the markup and marginal cost (Tirole, 1989). The prediction about how a marginal cost shock affects firm profitability follows from this model simple setup (all proofs in Appendix G).

Hypothesis 1 (firm profitability and marginal costs) *In simple models of imperfect competition, firm profitability declines following an increase in marginal costs.*

Consistent with the empirical evidence presented thus far on aggregate asset prices, the cost-shock hypothesis does indeed predict that investors should expect firm profitability to decline with inflation. Crucially, we also show that in the textbook model of imperfect competition, the key parameter for how much firm profits decline with marginal costs is a firm's market power.

Hypothesis 2 (market power, firm profitability, and marginal costs) *The extent to which firm profitability declines with marginal costs is determined by a firm's market power. In particular, firms with more market power see a smaller decline in profitability following an increase in marginal costs.*

This second hypothesis motivates the studying of the cross section of firms to help with understanding the economic drivers of stagflationary stock returns. In particular, we can exploit cross-sectional heterogeneity in firms' market power to ask whether firms with higher market power are expected to suffer smaller profitability declines following inflationary news, as the theory predicts. Cross-sectional empirical analysis will also allow us to estimate specifications that control for unobserved and observed time-varying factors affecting average stock returns and also to further narrow down the cash flow channel of stock returns.

In the results below, we first study the cross section of stock returns in response to inflationary news, before we then turn to investors' cash flow expectations. We initially extract cash flow expectations from observable discount rates and the return decomposition methodology. We then look at earnings expectations directly.

7.2 The Role of Market Power

We test the difference between firms with differential degrees of market power by estimating Equation 13. Figure IV plots the interaction coefficient between the inflationary news shock and markups. The interaction coefficient tests whether firms with higher markups exhibit differential stock returns around the announcement of inflationary news.

The results for $k < 0$ test for pre-trends. The close-to-zero and statistically insignificant coefficients before the CPI announcement suggest no violation of the parallel trends assumption, indicating that high- and low-markup firms would have experienced similar return dynamics absent the inflationary shock.

A positive interaction coefficient for $k \geq 0$ indicates that high-markup firms earn higher returns after inflationary news. Since markups are standardized (mean zero, standard deviation one), the coefficient represents the differential impact for firms with one standard deviation higher markup. The coefficient rises from 0.08 to 0.28 from the event day to five days after.³² Table VI presents results for $k = 5$ and confirms the results across various specifications with fixed effects and controls. For robustness, Table A.XIV replicates results using the Pellegrino (2025) cost elasticity measure, yielding qualitatively identical results.³³

Given that changes in discount rates affect the present value of discounted cash flows, differences in the cash flow duration of firms across the market power distribution would mechanically lead to heterogeneous returns in response to discount rate changes that occur with inflationary news. This variation in realized returns, which is purely due to discounting, would occur even if there was no change in expected cash flows across firms in the market power distribution.

To extract the cash flow component of variation in the cross section of stock returns, we therefore follow a similar two-stage strategy as in subsection 6.4.2. In the first stage, we estimate the following series of regressions for all $k \in [-5, 10]$ across CPI dates:

$$\widetilde{Return}_{i,t}^k = \alpha_i^k + (\Delta_k \widetilde{\mathbf{DCR}}_t' \times Markup_{i,y(t)-1}) \Theta^k + \Gamma_{i,t} \Psi^k + \widetilde{\epsilon}_{i,t}^k, \quad (14)$$

where $\widetilde{Return}_{i,t}^k$ is the real stock return around the CPI release and firm-level markups, $Markup_{i,y(t)-1}$, are interacted with a vector of discount rates, $\widetilde{\mathbf{DCR}}_t = [\widetilde{\mathbf{RF}}_t, ERP_t]$, that includes real risk-free yields across various maturities, $\widetilde{\mathbf{RF}}_t$, and the Martin (2017) equity risk premium, ERP_t . The regression also includes a control vector, $\Gamma_{i,t}$, that includes firm-level markups, $Markup_{i,t}$, the discount rate vector, $\widetilde{\mathbf{DCR}}_t$, other firm-level characteristics, $\mathbf{X}_{i,t}$: log assets, tangibility, leverage, book-to-market value, rolling betas to Fama and French (1993) asset pricing factors market beta, size, and value. The control vector $\Gamma_{i,t}$ additionally includes interactions between discount rates and firm-level characteristics.

³²This result is related to the post-announcement drift documented by Brooks et al. (2018) and Neuhierl and Weber (2018), and to the delayed adjustment documented by Altavilla et al. (2019) and Bauer et al. (2023) for macroeconomic and monetary-policy announcements.

³³We address identification concerns through matching and falsification tests. Using propensity score matching on firm characteristics (size, book-to-market, leverage, tangibility, factor loadings), we find the interaction coefficient remains stable or strengthens in matched samples (ranging from 0.219 to 0.305, all significant), with mean absolute standardized bias dropping from 11-15% to under 2.5% (Table A.XV). Placebo tests show the markup interaction is specific to CPI surprises ($t = 2.28$) and absent for employment ($|t| < 1.1$) and monetary policy surprises ($|t| < 1.1$), supporting our competition mechanism (Table A.XVI). Finally, controlling for sectoral price flexibility (Pasten et al., 2020, 2024), we find both channels operate independently: the markup interaction remains significant (0.165 to 0.177) when jointly estimated with price stickiness, though the latter is imprecisely estimated ($t \approx 1.25$) (Table A.XVII).

As discussed in [section 4](#) for the time series, by absorbing variation in returns related to changes in discount rates, we can then define the real cash flow component of firm-level stock returns around CPI releases as:

$$\widetilde{\epsilon}_{i,t}^k = \widehat{Return}_{i,t}^{k,CF}$$

and, because [Equation 14](#) does not include the inflation surprise itself in the equation, we can now test how this real cash flow component of stock returns respond differentially to inflationary news depending on the degree of market power firms have. To do so, we estimate in a second stage the following sequence of regressions for all $k \in [-5, 10]$ across CPI dates:

$$\begin{aligned} \widehat{Return}_{i,t}^{k,CF} = & \alpha + \beta_1 \text{Inflationary News}_t * \text{Low Markup}_{i,y(t)-1} \\ & + \beta_2 \text{Inflationary News}_t * \text{High Markup}_{i,y(t)-1} + \beta_3 \text{Low Markup}_{i,y(t)-1} + \epsilon_{i,t}, \end{aligned} \quad (15)$$

where $\widehat{Return}_{i,t}^{k,CF}$ is the predicted real cash flow component of the k -day stock returns. High markup firms are those firms that have at any given point in time markups above the 75th percentile of the markup distribution while those with low markups are those with markups below the 25th percentile of the distribution.

[Figure A5](#) plots β_1 and β_2 for different k 's. The left panel plots β_2 —the effect of inflationary news on the predicted real cash flow component of stock returns for high-markup firms. The coefficient is negative but not statistically significant for most of the horizon, indicating that investors do not expect real flows to decline with inflationary news for firms that do have market power. This result is consistent with the main model prediction that investors expect firms with market power to see nominal revenue decline less with inflation for firms with high market power if the inflation surprise is considered a cost shock. In contrast, for firms with a small degree of market power, investors expect real cash flows to decline significantly (right panel). With higher inflation, firms without market power see a stronger decline in profits.

In sum, inflationary news reduces stock prices for firms with limited market power, in large part driven by cash flow expectations, while high-markup firms are less severely affected. This suggests investors view inflation's impact on future cash flows more benignly for high-markup firms, consistent with inflation as a marginal cost shock ([Appendix G](#)).

Given that we find a real cash flow channel for the response of equity prices to inflationary news, a natural question is whether we observe a change in investors' expectations of company earnings. To tackle this question, we obtain changes in firm-level earnings expectations ($\Delta EarningsExpectation_{i,t}$) around CPI news, as described in [Appendix A](#). This variable allows us to test whether higher inflationary news are associated with different changes in earnings expectations across firms. To do so, we estimate the following empirical specification that is equivalent to [Equation 13](#) but replaces the firm-level return with the change in the firm-level earnings expect-

tations on the left-hand side:

$$\Delta EarningsExpectation_{i,t} = \alpha + \beta_1 \text{Inflationary News}_t * Markup_{i,y(t)-1} + \alpha_i + \alpha_t + \mathbf{X}'\gamma + \epsilon_{i,t}, \quad (16)$$

where, as previously, $\text{Inflationary News}_t$ is the difference between the published month-on-month CPI inflation and the median expectation of forecasts ahead of the announcement. $Markup_{i,y(t)-1}$ is the estimated markup from [De Loecker et al. \(2020\)](#) and standardized to have mean zero and a standard deviation of one. α_i is a firm fixed effect. α_t is a date fixed effect. $\mathbf{X}$ are controls and fixed effects, depending on the specification.

[Table VII](#) shows the results. In column (1) the coefficient on inflationary news shows the effect of the average firm, as markups are demeaned and standardized. Contrary to the idea that nominal earnings should increase with higher-than-expected inflation, we do not find evidence for the hypothesis that nominal earnings move higher with inflation. Instead, if anything, nominal earnings are expected to fall rather than increase after inflationary news. This result also suggests that earnings analysts do not see the inflationary surprise as a shock to demand, but rather as one to supply, which reduces earnings. The interaction coefficient between inflationary news and market power is positive and statistically significant, indicating that firms with market power are expected to experience smaller earnings declines than their counterparts. This result is consistent with the previous evidence that firms with market power earn higher stock returns because of higher expected earnings by the marginal stock market investor.

Columns (2)–(4) confirm the result when time fixed effects and other firm-level controls are included in the regression specification. Despite a substantial increase in the R-squared value, the coefficient estimate remains relatively stable, suggesting that market power, and not other firm characteristics that are in the control vector, is responsible for the differential response in earnings expectations around inflationary surprises.

8 Conclusion

In this paper, we have studied investors’ perceptions of inflation through the lens of a high-frequency event study. We have shown that inflation is expected to decrease firms’ real cash flows, which is consistent with interpreting inflation as stagflationary. These findings can be rationalized in a model in which inflation is seen as a shock to marginal costs, instead of one to demand. Another main prediction from this model is that market power shields firms’ earnings declines. When exploiting heterogeneity across firms, we have indeed found consistent evidence that stock prices, driven by differential earnings expectations, decline less for firms with more market power.

Overall, these results indicate that investors consider inflation bad news for the economy. Assuming investors are rational, and the lower real cash flows for firms are accurate, one may want to conclude that surprise inflation does not only lead to stagflationary stock returns but also to stagflation in the macroeconomy. However, drawing implications from the movement in cash flow

expectations for the actual implications of inflation should be done with caution. While stock returns may have real effects themselves (Bond et al., 2012), investors' expectations may contain systematic errors, and their expectations are not in line with realized future economic outcomes (Bordalo et al., 2024).

Hence, our results raise important further questions, which we leave for future research. Is investors' perception that inflationary news are stagflationary accurate? Or does Wall Street have a biased "stagflationary" view of the world, as it incorrectly interprets inflationary news as supply shocks that drive up marginal costs? Does higher inflation lead to an increasing market share of firms with already high market power, potentially leading to higher prices in aggregate? What are the implications for monetary policy if an initial increase in inflation leads to more inflation because of an aggregate increase in market power?

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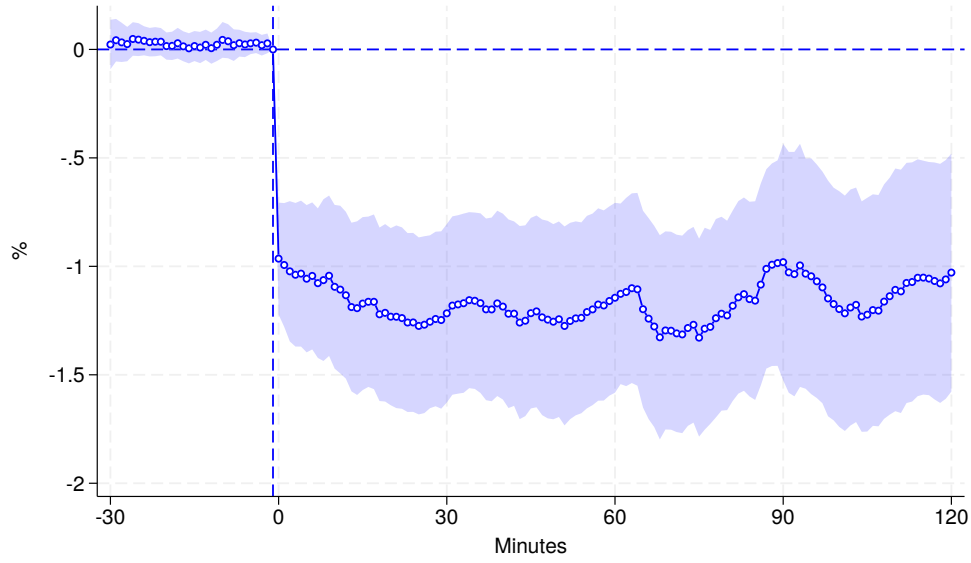
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Figure I: Stock returns around inflationary news

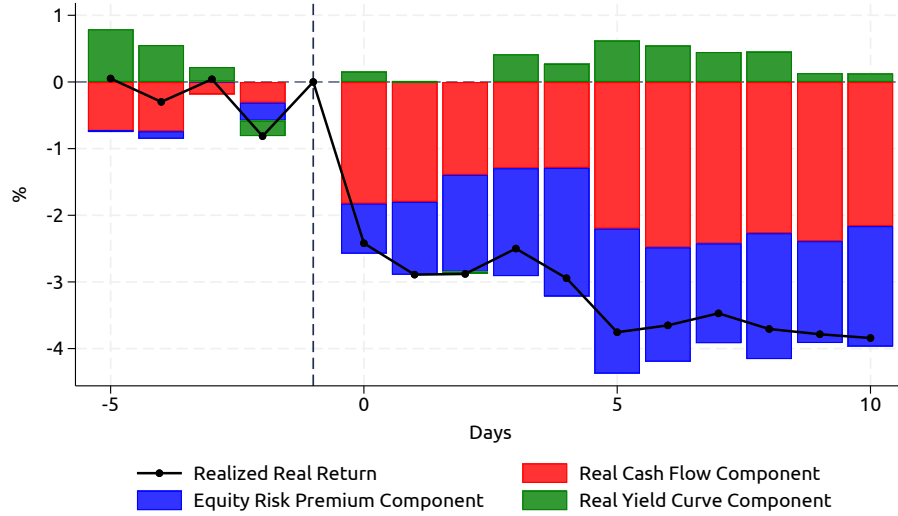


This figure plots the estimated β^k coefficient for the following sequence of regressions for all $k \in [-30, 120]$ across CPI dates:

$$y_t^k = \alpha^k + \beta^k \text{Inflationary News}_t + \epsilon_t^k,$$

where y_t^k is the log difference in the Emini S&P500 futures price from the minute before the CPI release to k minutes before (if $k < 0$) or k minutes after (if $k > 0$) the CPI release. The release time is 8:30am in all observations. $\text{Inflationary News}_t$ is the difference between the published month-on-month CPI inflation and the median expectation of forecasts ahead of the announcement. The sample period is 1997–2022 and the shaded area reflects the 90% confidence interval using robust standard errors.

Figure II: Decomposition of real stock returns around inflationary news

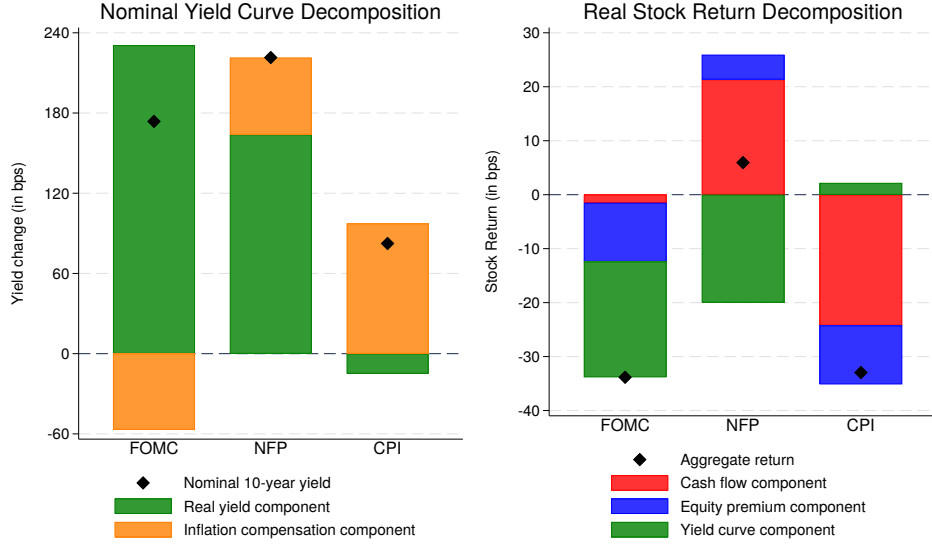


This figure stacks the estimated coefficient of Equation 12:

$$y_t^{k,c} = \alpha^{k,c} + \beta^{k,c} \text{Inflationary News}_t + \epsilon_t^{k,c},$$

where $y_t^{k,c}$ is the predicted $c = \{\widetilde{CF}, \widetilde{EP}, \widetilde{YC}\}$ component of the cumulative real return between the day before the CPI announcement on date t and k days after. The return components are the returns generated from changes in real cash flow expectations, $\widetilde{CF}$, changes in equity risk premium, $\widetilde{EP}$, and changes in real government bond yields, $\widetilde{YC}$. The black line in the figure plots the coefficients of real realized return regressed on inflation news for $k = -5$ through $k = 10$, which is by definition the sum of the stacked coefficients on the return components. Inflationary News $_t$ is the difference between the published month-on-month CPI inflation and the median expectation of forecasts ahead of the announcement. The sample period is 1999–2022.

Figure III: Nominal yield and stock return decomposition around different macroeconomic news

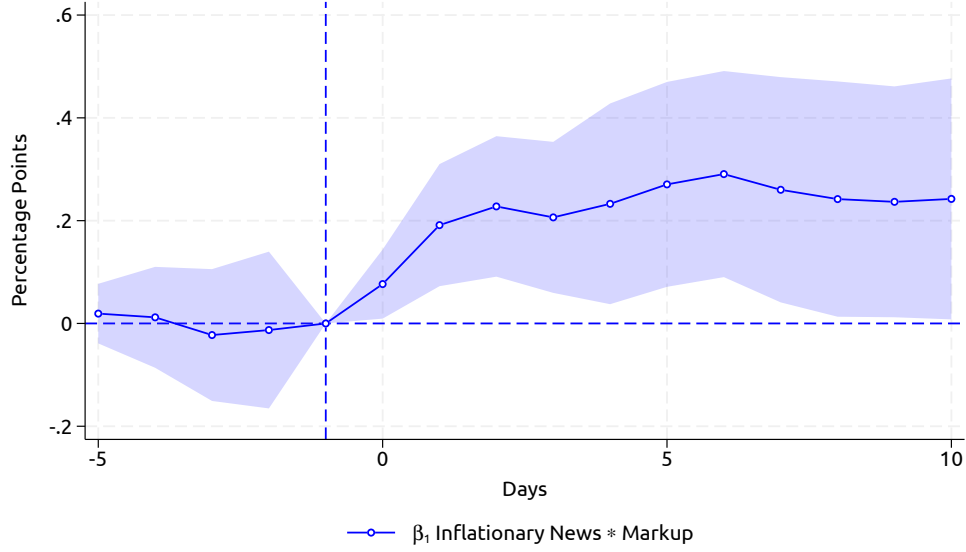


The left figure stacks the estimated coefficient of the one-day change in real yield and inflation compensation components of the 10-year nominal yield change regressed on news that day. The black line in the figure plots the coefficients of the nominal yield regressed on the news released that day, which is by definition the sum of the stacked coefficients on the return components. The right figure stacks the estimated coefficient of the equation:

$$y_t^c = \alpha^c + \beta^c \text{News}_t + \epsilon_t^c,$$

where y_t^c is the $c = \{CF, EP, YC\}$ component of the one-day return on the day t of a FOMC, NFP or CPI announcement. The return components are the returns generated from changes in real cash flow expectations, CF , changes in equity premium, EP , and changes in government bond real yields, YC . The black line in the figure plots the coefficients of the total realized return regressed on the news released that day, which is by definition the sum of the stacked coefficients on the return components. FOMC news is the [Bauer and Swanson \(2023\)](#) high-frequency monetary policy surprise, and NFP and CPI news are the differences between actual data releases and median forecaster expectations. All news are standardized.

Figure IV: The role of market power for stock returns around inflationary news



This figure plots the estimated coefficient of Equation 13:

$$Return_{i,t}^k = \alpha + \beta_1 \text{Inflationary News}_t * Markup_{i,y(t)-1} + \alpha_i + \alpha_t + \mathbf{X}'\gamma + \epsilon_{i,t}$$

where $Return_{i,t}^k$ is the cumulative return between the day before the CPI announcement on date t and k days after for stock i . $\text{Inflationary News}_t$ is the difference between the published month-on-month CPI inflation and the median expectation of forecasts ahead of the announcement. $Markup_{i,y(t)-1}$ is the estimated markup from De Loecker et al. (2020) and standardized to have mean 0 and a standard deviation of 1. α_i is a firm fixed effect. α_t is a date fixed effect. $\mathbf{X}$ are controls. Standard errors are double clustered at the firm and date level. The shaded area reflects the 90% confidence interval.

Table I: High-frequency asset price changes around inflationary news

Panel A: OLS estimation					
	(1) Equity return	(2) 2-year nominal yield	(3) 10-year nominal yield	(4) VIX	(5) 2-year inflation swap
Inflationary News	-1.121*** (0.214)	6.647*** (1.664)	5.749*** (1.041)	2.340*** (0.713)	30.996*** (5.076)
R-squared	0.178	0.095	0.139	0.076	0.324
N	251	251	251	251	94

Panel B: Heteroskedasticity-based Kalman filter estimation					
	(1) Equity return	(2) 2-year nominal yield	(3) 10-year nominal yield	(4) VIX	(5) 2-year inflation swap
Headline News	-1.122*** (0.146)	6.653*** (1.289)	5.748*** (0.922)	2.339*** (0.409)	26.958*** (4.403)
Latent News	-0.259*** (0.016)	2.218*** (0.147)	1.526*** (0.107)	0.633*** (0.054)	2.284*** (0.542)
R-squared	0.763	0.697	0.667	0.592	0.434

This table shows asset price changes around inflationary news in narrowly measured high-frequency windows. Panel A reports OLS estimation results from the following equation:

$$y_t = \alpha + \beta \text{Inflationary News}_t + \epsilon_t,$$

where y_t is the 10-minute change in an asset price from 5 minutes before to 5 minutes after Inflationary News_{*t*}, which is the difference between the published month-on-month CPI inflation and the median expectation of forecasts ahead of the announcement. In column (5), y_t is the 95 minute change in the inflation swap from 5 minutes before the inflationary news to 90 after. The estimated constant coefficients are omitted from the table, and robust standard errors are reported in parentheses. Panel B shows [Gurkaynak et al. \(2020\)](#) maximum-likelihood estimates from the following equation:

$$y_t = \beta \text{Inflationary News}_t + \gamma d_t f_t + \epsilon_t,$$

where $d_t = 1$ on announcement windows and 0 on the matched non-announcement windows sampled at the same time but on nearby non-CPI release days, and f_t is a scalar latent news factor realized only on announcement windows. A maximum-likelihood estimate of the latent-factor model is estimated jointly across the four market assets in columns 1-4. The reported R^2 is the model-implied share of announcement-window variance attributable to the systematic (headline plus latent) components. Standard errors in parentheses are from the observed information matrix. Column 5 results present an OLS regression of the inflation swap change on the headline inflationary surprise and the latent factor estimated in the other four asset markets. In both panels, columns 1 and 4 present asset price returns (log changes measured in percent), and columns 2, 3 and 5 present yield changes measured in basis points. The sample period is 1997–2022 in columns 1 to 4 and 2007–2022 in column 5, with observations where there are fewer than 20 unique asset price changes on a given CPI announcement day removed. *** p<0.01, ** p<0.05, * p<0.1.

Table II: Daily asset price changes around inflationary news

Panel A: 1-day asset price changes

	Equity Return		2yr nominal yield		10yr nominal yield		1yr equity premium	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Inflationary News	-1.04** (0.41)	-0.92** (0.40)	7.95*** (2.24)	7.68*** (2.26)	8.16*** (2.38)	8.21*** (2.45)	31.78* (19.02)	23.13 (14.67)
Controls		✓		✓		✓		✓
R-squared	0.021	0.073	0.028	0.069	0.028	0.070	0.017	0.121
N	536	536	536	536	536	536	438	438

Panel B: 5-day asset price changes

	Equity Return		2yr nominal yield		10yr nominal yield		1yr equity premium	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Inflationary News	-1.89** (0.81)	-1.43* (0.82)	1.87 (7.04)	0.51 (7.55)	6.59 (6.33)	7.04 (6.60)	61.94* (33.62)	59.49* (35.48)
Controls		✓		✓		✓		✓
R-squared	0.010	0.101	0.000	0.044	0.003	0.040	0.011	0.132
N	536	536	536	536	536	536	438	438

This table shows estimation results from the following equation:

$$y_t = \alpha + \beta \text{Inflationary News}_t + \mathbf{X}'_t \gamma + \epsilon_t,$$

where y_t is the change in an asset price measured from the price at the close of business the day before Inflationary News_{*t*} is released to the price at the close of business the day of (panel A) or 5 business days after (panel B) the Inflationary News_{*t*} is released. $\mathbf{X}_t$ is a vector of controls where each element is the surprise over the event window for another type of macroeconomic or monetary surprise. The vector includes 22 other macroeconomic data releases as well as [Bauer and Swanson \(2023\)](#) monetary policy surprises with FOMC statements and press conferences separated into two elements of the vector. The estimated constant coefficients are not reported. Robust standard errors are reported in parentheses. The sample period is 1977-2022. *** p<0.01, ** p<0.05, * p<0.1.

Table III: Decomposing the yield curve around inflationary news

Panel A: 2-year maturity						
	nominal yield		inflation comp		real yield	
	(1)	(2)	(3)	(4)	(5)	(6)
Inflationary News	9.05*** (2.99)	8.55*** (2.98)	34.66*** (7.08)	37.91*** (7.34)	-25.61*** (7.40)	-29.37*** (7.35)
Controls		✓		✓		✓
R-squared	0.044	0.130	0.066	0.117	0.036	0.109
N	283	283	283	283	283	283

Panel B: 10-year maturity						
	nominal yield		inflation comp		real yield	
	(1)	(2)	(3)	(4)	(5)	(6)
Inflationary News	6.07* (3.32)	6.12* (3.45)	7.00*** (2.53)	8.49*** (2.61)	-0.94 (3.06)	-2.38 (2.85)
Controls		✓		✓		✓
R-squared	0.015	0.092	0.040	0.149	0.000	0.192
N	283	283	283	283	283	283

This table shows estimation results from the following equation:

$$y_t = \alpha + \beta \text{Inflationary News}_t + \mathbf{X}_t' \gamma + \epsilon_t,$$

where y_t is the 1-day change in yield around Inflationary News $_t$ that is released at 8:30am on day t . Yield changes are measured in basis points. $\mathbf{X}_t$ is a vector of controls where each element is the surprises over the event window for another type of macroeconomic or monetary surprise. The vector includes 22 other macroeconomic data releases as well as [Bauer and Swanson \(2023\)](#) monetary policy surprises with FOMC statements and press conferences separated into two elements of the vector. Panel A reports results for 2-year yields and panel B reports results for 10-year yields. The estimated constant coefficients are not reported. Robust standard errors are reported in parentheses. The sample period is 1999-2022. *** p<0.01, ** p<0.05, * p<0.1.

Table IV: Decomposing stock returns around inflationary news

	Nominal Decomposition				Real Decomposition			
	(1) Return	(2) Yield Curve	(3) ERP	(4) Cash Flow	(5) Return	(6) Yield Curve	(7) ERP	(8) Cash Flow
Inflationary News	-1.420* (0.847)	-0.612 (0.462)	-1.049 (0.942)	0.241 (0.597)	-2.420*** (0.847)	0.155 (0.470)	-0.749 (0.727)	-1.825*** (0.594)
R-squared	0.021	0.008	0.013	0.001	0.059	0.001	0.012	0.045
N	283	283	283	283	283	283	283	283

The table presents the coefficients from the following equation:

$$y_t = \alpha + \beta \text{Inflationary News}_t + \epsilon_t$$

where y_t is the one-day return on a component of the stock market from the close of business the day before Inflationary News_{*t*} is released to the close of business the day of the release. Columns 1 and 5 present estimation results where y_t is the total market return and columns 2-4 and 6-8 present estimation results where y_t is the yield curve, equity premium or cash flow components of the return. Columns 1-4 show coefficient estimates for the nominal return decomposition and columns 5-8 show coefficient estimates for the real return decomposition. Inflationary News_{*t*} is the difference between the published month-on-month CPI inflation and the median expectation of forecasts ahead of the announcement. Constant parameter estimates are unreported. Standard errors in parentheses are based on 2,000 bootstrap replications. In each bootstrap sample, we re-estimate the return decomposition and regenerate the ERP and cash-flow components, then re-estimate the reported regressions. *** p<0.01, ** p<0.05, * p<0.1. The sample period is 1999–2022.

Table V: Variation in stock return decomposition results across periods

Panel A: Nominal return decomposition				
	Return	Yield Curve	ERP	Cash Flow
Inflationary News	-0.82 (0.66)	-0.53 (0.47)	-0.81 (0.66)	0.53 (0.58)
GFC recession	0.01* (0.00)	0.00* (0.00)	0.00 (0.00)	-0.00 (0.00)
Elevated Oil Volatility	-0.00 (0.00)	0.00 (0.00)	-0.00 (0.00)	0.00 (0.00)
Inflationary News x GFC	3.14** (1.54)	-2.67** (1.11)	4.18*** (1.54)	1.63 (1.37)
Inflationary News x Oil	-4.82*** (1.37)	1.50 (0.99)	-4.30*** (1.37)	-2.02* (1.22)
R-squared	0.089	0.041	0.072	0.015
N	282	282	282	282

Panel B: Real return decomposition				
	Return	Yield Curve	ERP	Cash Flow
Inflationary News	-1.82*** (0.64)	0.35 (0.42)	-0.57 (0.51)	-1.60*** (0.52)
GFC recession	0.01 (0.01)	0.00 (0.00)	0.00 (0.01)	0.00 (0.01)
Elevated Oil Volatility	-0.00 (0.00)	0.00 (0.00)	-0.00 (0.00)	-0.00 (0.00)
Inflationary News x GFC	3.14 (3.00)	-1.22 (1.88)	2.87 (3.48)	1.49 (2.75)
Inflationary News x Oil	-4.82** (2.28)	-0.03 (1.39)	-3.11 (2.15)	-1.69 (2.01)
R-squared	0.124	0.015	0.067	0.052
N	282	282	282	282

The table presents the coefficients from the following equation:

$$Return_t^c = \alpha^c + \beta^c \text{Inflationary News}_t + \mathbf{I}_{t=p}' \cdot \Gamma_p^c + (\text{Inflationary News}_t \times \mathbf{I}_{t=p}') \cdot \Gamma_{I \times p}^c + \epsilon_t,$$

where $Return_t^c$ is the 1-day return for component $c = \{\text{total, yield curve, equity risk premium, cash flow}\}$ of the stock market return. $\text{Inflationary News}_t$ is the difference between the published month-on-month CPI inflation and the median expectation of forecasts ahead of the announcement. $\mathbf{I}_{t=p}$ is a vector of indicator variables set to 1 if date t falls within period p . There are two periods included in the vector: the global financial crisis recession (defined by NBER as December 2007 through to June 2009) and months that fall in the top quartile of realized oil price volatility. Panel A presents a nominal return decomposition and panel B presents a real return decomposition. The sample period is 1999–2022. Standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table VI: Inflationary news, market power, and stock returns

	Dependent Variable: $Return_{i,t}^5$					
	(1)	(2)	(3)	(4)	(5)	(6)
Inflationary News	-1.501* (0.798)					
Markup	0.0129 (0.0147)	0.0105 (0.0150)	-0.000807 (0.00925)	0.00674 (0.00938)	-0.00147 (0.0101)	0.00545 (0.0103)
Inflationary News $\times$ Markup	0.260** (0.115)	0.271** (0.121)	0.165** (0.0805)	0.174** (0.0755)	0.185** (0.0838)	0.177** (0.0804)
R-squared	0.014	0.128	0.155	0.156	0.165	0.165
N	2,283,661	2,283,661	2,283,628	2,227,067	1,651,423	1,628,093
Firm FE	✓	✓	✓	✓	✓	✓
Time FE		✓	-	-	-	-
Industry-Time FE			✓	✓	✓	✓
Int. Firm Controls				✓		✓
Int. Factor Controls					✓	✓

This table shows results from Equation 13:

$$Return_{i,t}^5 = \alpha + \beta_1 \text{Inflationary News}_t * \text{Markup}_{i,y(t)-1} + \alpha_i + \alpha_t + \mathbf{X}'\gamma + \epsilon_{i,t},$$

where $Return_{i,t}^5$ is the cumulative stock returns calculated from day $t-1$, before CPI release, to day $t+5$, after the CPI release. $\text{Inflationary News}_t$ is the difference between the published month-on-month CPI inflation and the median expectation of forecasts ahead of the announcement. $\text{Markup}_{i,y(t)-1}$ is the estimated markup from De Loecker et al. (2020) and is standardized to have mean 0 and a standard deviation of 1. α_i is a firm fixed effect. α_t is a date fixed effect. $\mathbf{X}$ are controls. *Int. Firm Controls* includes firm characteristics controls—log assets, tangibility, leverage, and market-to-book value—interacted with Inflationary News. *Int. Factor Controls* includes firm-level rolling-betas to the Fama and French (1993) asset pricing factors—market beta, size, and value, each interacted with Inflationary News. The sample period is 1977–2022. Standard errors (in parentheses) are double clustered at the firm and date level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table VII: Inflationary news, market power, and earnings expectations

	Dependent Variable: $\Delta EarningsExpectation_{i,t}$			
	(1)	(2)	(3)	(4)
Inflationary News	-0.313 (0.828)	-0.330 (0.834)		
Markup	0.019 (0.052)	-0.108 (0.096)	-0.075 (0.096)	-0.054 (0.099)
Inflationary News $\times$ Markup	0.842* (0.471)	0.910* (0.475)	1.140** (0.480)	1.182** (0.489)
R-squared	0.000	0.026	0.054	0.054
N	44,627	44,603	44,602	42,898
Firm FE		✓	✓	✓
Time FE			✓	✓
Int. Firm Controls				✓

This table shows results from [Equation 16](#):

$$\Delta EarningsExpectation_{i,t} = \alpha + \beta_1 \text{Inflationary News}_t * \text{Markup}_{i,y(t)-1} + \alpha_i + \alpha_t + \mathbf{X}'\gamma + \epsilon_{i,t},$$

where $\Delta EarningsExpectation_{i,t}$ is the log difference in earnings expectations for firms between the 15 days after and the 15 days before the CPI announcement at date t . $\text{Inflationary News}_t$ is the difference between the published month-on-month CPI inflation and the median expectation of forecasts ahead of the announcement. $\text{Markup}_{i,y(t)-1}$ is the estimated markup from [De Loecker et al. \(2020\)](#) and standardized to have mean 0 and a standard deviation of 1. α_i is a firm fixed effect. α_t is a date fixed effect. $\mathbf{X}$ are controls. *Int. Firm Controls* includes firm characteristics controls—log assets, tangibility, leverage, and market-to-book value—interacted with Inflationary News. Standard errors (in parentheses) are double clustered at the firm and date level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Internet Appendix to “Stagflationary Stock Returns”

This Appendix provides additional descriptive and empirical evidence to supplement the analyses provided in the main text. It also provides a simple model that motivates the cross-sectional empirical analysis. Below, we list the content.

1. [Appendix A](#) describes the details of the construction of the earnings expectations dataset .
2. [Appendix B](#) describes the details of the construction of the SPF dataset.
3. [Appendix C](#) reports the response of dividend futures to inflationary news.
4. [Appendix D](#) presents the results of a VAR-decomposition around inflationary news.
5. [Appendix E](#) studies the interaction of long-dated interest rates with short-dated nominal rates around inflationary news using the empirical setting of [Hanson and Stein \(2015\)](#).
6. [Appendix F](#) studies how the response of the stock market to inflation news depends on the state of the economy.
7. [Appendix G](#) presents a simple model with imperfect competition and studies the impact of changing marginal costs for firm profitability in this setting.
8. [Table A.VIII](#) reports summary statistics of the variables used in the main empirical analysis.
9. [Table A.IX](#) demonstrates how changes in inflation expectations around CPI releases affect stock returns.
10. [Table A.X](#) studies the role of core inflation shocks.
11. [Table A.XI](#) decomposes the driver of the equity risk premium response.
12. [Table A.XII](#) shows the response of inflation swaps to inflationary news.
13. [Table A.XIII](#) shows the perceived comovement of revisions in SPF inflation and growth expectations.
14. [Table A.XIV](#) studies the interaction between inflationary news, cost elasticity, and stock returns.
15. [Figure A2](#) reports the response of forward nominal yields to inflation news.
16. [Figure A3](#) presents the validation exercise showing the correlation of our cash flow measures with direct measures.
17. [Figure A4](#) presents the validation exercise showing earnings and cash flow expectations in response to NFP and inflationary surprises.
18. [Figure A5](#) presents the real estimated cash flow response by market power in response to inflationary news.

A Construction of Analyst Earnings Expectations

We obtain analyst earnings forecasts from the Thomson Reuters IBES Estimates Database. Following the literature, we retain only firms that are constituents of the S&P 500, as earnings forecasts for these firms are generally based on broader analyst coverage and are therefore likely to be more informative. Individual earnings-per-share forecasts are converted into earnings forecasts by multiplying earnings per share by shares outstanding.

For each analyst forecast of firm i reported by analyst a on date d , we identify the closest CPI release date t and compute the distance between the forecast date and the announcement date. We retain only forecasts made within a 15-day window around the CPI release, i.e.,

$$|d - t| \leq 15.$$

For each firm and CPI announcement, we separately average analyst forecasts submitted before and after the announcement,

$$Earnings\ Expectation_{i,t}^{pre} = \frac{1}{N_{i,t}^{pre}} \sum_{d-t < 0} Earnings\ Expectation_{i,a,d,t},$$

and

$$Earnings\ Expectation_{i,t}^{post} = \frac{1}{N_{i,t}^{post}} \sum_{d-t \geq 0} Earnings\ Expectation_{i,a,d,t},$$

where $N_{i,t}^{pre}$ and $N_{i,t}^{post}$ denote the number of analyst forecasts available for firm i around CPI announcement t pre and post the event.

We measure the revision in earnings expectations around the CPI announcement as

$$\Delta EarningsExpectation_{i,t} = \log(Earnings\ Expectation_{i,t}^{post}) - \log(Earnings\ Expectation_{i,t}^{pre}).$$

The firm-level revisions are used throughout the cross-sectional analysis. To obtain an aggregate time-series measure comparable to our market-based expectations, we compute the asset-weighted average of firm-level earnings revisions across all firms for each CPI announcement.

Summary statistics are reported in Appendix Table A.I. On average, approximately three analysts cover each firm before and after a CPI release, corresponding to roughly 600–700 analyst forecasts around a typical announcement.

Table A.I: Summary statistics—earnings expectations

	N	Mean	SD	p10	p25	p50	p75	p90
Analysts per firm before news	381	3.4	1.8	1.6	2	3	4	6.1
Analysts per firm after news	381	3.5	2.6	1.5	1.9	2.4	4	8.1
Analysts total before news	381	711	861	100	190	344	668	2177
Analysts total after news	381	587	486	113	234	417	849	1337
$\Delta AverageEarningsExpectation$	381	.31	4.4	-4.1	-1.8	.13	2.6	4.8

This table presents summary statistics for the earnings expectations analysis.

B Construction of Survey of Professional Forecasters Variables

We obtain individual-level forecasts from the Survey of Professional Forecasters (SPF). Specifically, we use four-quarter-ahead forecasts of CPI inflation and real GDP from individual respondents. The panel is organized at the forecaster-quarter level.

To measure how beliefs respond to macroeconomic news, we construct quarter-to-quarter forecast revisions for each forecaster,

$$\Delta\pi_{i,t}^e = \pi_{i,t}^e - \pi_{i,t-1}^e,$$

and

$$\Delta g_{i,t}^e = g_{i,t}^e - g_{i,t-1}^e,$$

where $\pi_{i,t}^e$ denotes forecaster i 's four-quarter-ahead expected CPI inflation and $g_{i,t}^e$ denotes the corresponding four-quarter-ahead expected annualized real GDP growth.

Using these revisions, we estimate the perceived Phillips curve from the regression

$$\Delta\pi_{i,t}^e = \alpha_i + \alpha_t + \beta\Delta g_{i,t}^e + \varepsilon_{i,t},$$

where α_i and α_t denote forecaster and quarter fixed effects. To examine whether the perceived trade-off varies with the macroeconomic environment, we augment the specification with interactions between expected growth revisions and indicators for the Global Financial Crisis and lagged oil-price volatility. Standard errors are two-way clustered by forecaster and quarter.

The estimated coefficient β measures the perceived Phillips curve embedded in professional forecasts. A positive coefficient indicates that forecasters revise inflation and growth expectations in the same direction, consistent with a demand-driven interpretation of macroeconomic news, whereas a flat or negative coefficient indicates that inflation revisions are largely disconnected from growth revisions, consistent with a supply-driven interpretation.

C Dividend Future Prices around Inflationary News

In this section we consider how dividend futures respond to inflationary news in the later sample where we observe dividend future prices (2016–2022). Panel A of [Table A.II](#) presents log changes in the constant-maturity 1-year and 2-year dividend futures price around inflation news. It presents 1-day changes and 5-days following inflation announcements and for each window presents both nominal and real changes. For real changes, dividend future prices are discounted by expected inflation of the same maturity (as measured from breakeven inflation rates) so that they are adjusted from nominal to real quantities, and changes are adjusted for the inflationary news that reflects an immediate change in the economy price level. Panel A of [Table A.II](#) shows that nominal dividend futures are close to unchanged following inflationary news, while real cash flows exhibit statistically significant declines. For example, following a 1 percentage point inflationary surprise, the 1-year dividend future declines by 162 basis points that day on a real basis. The results from dividend futures are consistent with the previous analysis extracting cash flow news from the decomposition approach.

As dividend futures are risk neutral-expectations of stock market dividend payments their price declines following inflationary news could partially reflect increases in risk premium. Following [Knox and Vissing-Jorgensen \(2024\)](#), we therefore adjust dividend futures prices for the [Martin \(2017\)](#) lower bound of equity risk premium at the same maturity, which generates a lower bound of expected dividends. The results of regressing these adjusted dividend futures prices on inflationary news are presented in [Table A.II](#) of Panel B, with the key takeaways being broadly the same as in Panel A: Nominal cash flow expectations are unchanged following inflationary news, while real dividend expectations decline significantly.

Table A.II: Dividend futures, expected dividends, and inflationary news

Panel A: Dividend Futures

	1-day log change				5-day log change			
	1-yr nom	2-yr nom	1-yr real	2-yr real	1-yr nom	2-yr nom	1-yr real	2-yr real
Inflationary News	-0.09 (0.14)	-0.17 (0.32)	-1.62*** (0.15)	-1.85*** (0.31)	-0.70 (1.17)	-1.72 (2.44)	-2.09* (1.11)	-3.10 (2.35)
R-squared	0.003	0.004	0.511	0.312	0.002	0.002	0.016	0.008
N	77	77	77	75	76	76	75	74

Panel B: Lower Bound of Expected Dividend

	1-day log change				5-day log change			
	1-yr nom	2-yr nom	1-yr real	2-yr real	1-yr nom	2-yr nom	1-yr real	2-yr real
Inflationary News	0.15 (0.17)	0.06 (0.21)	-1.38*** (0.20)	-1.62*** (0.25)	0.15 (0.77)	-0.83 (1.94)	-1.26* (0.74)	-2.21 (1.86)
R-squared	0.010	0.001	0.441	0.354	0.000	0.001	0.015	0.007
N	77	77	77	75	76	76	75	74

This table shows estimation coefficients from the regression:

$$\Delta_k \ln(x_t^n) = \alpha_k + \beta_k \text{Inflationary News}_t + \epsilon_t^k,$$

where the dependent variable x_t^n is the $n = \{1, 2\}$ -year dividend futures price (Panel A) or lower bound of the expected dividend (Panel B). The expected dividend is the dividend futures prices adjusted for risk premium (as measured by the [Martin \(2017\)](#) lower bound of the equity risk premium). Panel A and Panel B both present log changes in nominal and in real terms. The real versions adjusts nominal changes for changes in expected inflation (measured by inflation swap rates) and the surprise component of realized inflation over the change period Δ_k . The regressions are estimated over the sample period 2016–2022 and robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

D Cash Flows Implied by a VAR Methodology

In this section we follow the method of [Bernanke and Kuttner \(2005\)](#), the seminal application of the [Campbell and Ammer \(1993\)](#) VAR decomposition, that conditions on macroeconomic announcement news. This approach allows us to recover the VAR-implied cash-flow, real-rate, and equity-premium components of the stock market's response to inflationary news.

D.1 Methodology

By the [Campbell and Shiller \(1988\)](#) log-linearization, an unexpected real log market return must reflect a revision in expectations about either future cash flows or future discount rates. Splitting the discount rate into a real-rate and an excess-return (equity-premium) piece gives the three-way identity

$$\tilde{h}_t = \tilde{\eta}_{d,t} - \tilde{\eta}_{r,t} - \tilde{\eta}_{e,t} \quad (17)$$

Writing r for the real interest rate, and d for log dividends, the three news terms are the revisions in the discounted present values of, respectively, future dividend growth, future real rates, and future excess returns:

$$\tilde{\eta}_{d,t} = (E_t - E_{t-1}) \sum_{j=0}^{\infty} \rho^j \Delta d_{t+j}, \quad \tilde{\eta}_{r,t} = (E_t - E_{t-1}) \sum_{j=1}^{\infty} \rho^j r_{t+j}, \quad \tilde{\eta}_{e,t} = (E_t - E_{t-1}) \sum_{j=1}^{\infty} \rho^j (h_{t+j} - r_{t+j}), \quad (18)$$

where $h - r$ is the excess return and ρ is the linearization constant. Equation (17) is an accounting identity for expectations: a return surprise not accounted for by revised equity premium ($\tilde{\eta}_e$) or real-rate ($\tilde{\eta}_r$) expectations must be attributed to a revised cash flow ($\tilde{\eta}_d$),

To estimate the expectation revisions in (17), we follow [Campbell and Ammer \(1993\)](#) and estimate them with a 6 element first-order VAR whose first element is the real return:

$$z_t = \mu + A z_{t-1} + w_t, \quad z_t = [h_t, r_t, \Delta b_t, s_t, dp_t, rb_t]', \quad (19)$$

with h_t the real log return, r_t the real bill rate, Δb_t the change in the nominal bill rate, s_t the term spread, dp_t the dividend-price ratio, and rb_t the relative bill rate. With the linearization constant $\rho = 0.9962$ and $\Lambda \equiv \rho A (I - \rho A)^{-1}$, and letting e_1 and e_2 select the real return and the real rate from z_t , the news terms are linear in the VAR innovation w_t :

$$\tilde{\eta}_{r,t} = e_2' \Lambda w_t, \quad \tilde{\eta}_{e,t} = (e_1 - e_2)' \Lambda w_t, \quad \tilde{\eta}_{d,t} = \tilde{h}_t + \tilde{\eta}_{r,t} + \tilde{\eta}_{e,t}, \quad (20)$$

with $\tilde{h}_t = e_1' w_t$. Real-rate and excess-return news are forecast directly from the VAR dynamics. As is standard in the literature, the dividend process is therefore not modeled with cash-flow news backed out as the residual that enforces (17).

To measure the response of the real return components to inflationary news, we condition the decomposition on the CPI surprises that are used in our baseline analysis. Following [Bernanke and Kuttner \(2005\)](#), the surprise enters the VAR as an exogenous regressor,

$$z_t = \mu + A z_{t-1} + \phi \text{Inflationary News}_t + w_t^\perp, \quad (21)$$

so that ϕ is the contemporaneous (impact) response of each state variable to a unit surprise and $w_t^\perp$ is orthogonal to it. We estimate (21) in two steps: first the dynamics A from (19), then ϕ by regressing the VAR’s one-step-ahead residuals on Inflationary News_{*t*} equation by equation. Substituting the estimated ϕ for w_t in (20) gives the response of each return component to the inflation surprise.

D.2 Results

Table A.III presents the estimated first-order VAR coefficient matrix. Each column reports the OLS estimates of one equation (the dependent variable named in the column head regressed on the full lagged state z_{t-1}). These reduced-form VAR coefficients and R^2 are identical equation by equation to a system estimator.

As is well-established in the literature, the dividend–yield is the crucial forecasting variable for the VAR-implied equity premium return component. In column (1), we see that a higher dividend price ratio (i.e. low valuations) forecasts real returns, while in column 5 we see that the dividend yield is highly persistent with a loading on its own lag of 0.994. This means that even small movements in the dividend yield lead to durable revisions in future expected return that compound through Λ into a large equity price move.

Table A.IV presents the variance decomposition of the unexpected real market return implied by the estimated VAR. From the identity $\tilde{h}_t = \tilde{\eta}_{d,t} - \tilde{\eta}_{r,t} - \tilde{\eta}_{e,t}$, the variance satisfies

$$\text{Var}(\tilde{h}) = \text{Var}(\tilde{\eta}_d) + \text{Var}(\tilde{\eta}_r) + \text{Var}(\tilde{\eta}_e) - 2 \text{Cov}(\tilde{\eta}_d, \tilde{\eta}_r) - 2 \text{Cov}(\tilde{\eta}_d, \tilde{\eta}_e) + 2 \text{Cov}(\tilde{\eta}_r, \tilde{\eta}_e).$$

Consistent with the literature, we find a large role for the excess-return news and the covariance of excess-returns with cash-flow news. Interestingly, over our sample period (1977-2022), the role of cash-flow news is somewhat larger than in the seminal Campbell and Ammer (1993).

Table A.V presents the real return response to inflationary news decomposed into news components. We find that a positive CPI surprise lowers expected future cash flows, and this channel accounts for the largest share of the roughly one-percentage-point negative real return. This finding is consistent with our main methodology, although the remaining decomposition between the two discount rate components is somewhat different, with the VAR suggesting a smaller role for equity risk premium.

D.3 Limitations

We are wary, however, of over-interpreting the conditional VAR decomposition. Our first caution is the concern that Nagel and Xu (2024) articulate. As the authors describe, the fact that the dividend–price ratio (D/P) is generally associated with movements in expected excess returns, as captured in the unconditional VAR, does not necessarily mean that its movements around a selected set of specific events share that same composition. The VAR delivers an average association across all the reasons D/P moves. However, conditional on one particular type of news (i.e., monetary or inflationary), the same D/P movement could instead reflect changing expectations of future real rates or changing cashflow expectations.

A second concern is that the VAR is estimated at a monthly frequency given the data availability

of the underlying state variables, which dramatically reduces power from estimating the effects of inflationary surprises relative to our high-frequency event-study setting. Through the lens of the VAR, we find unexpected real returns at the monthly frequency fall roughly one percent with the within-month inflationary news. Reassuringly, this is in the same direction and quantitatively similar to realized return measured in high frequency. However, the r-square is 0.1 percent and estimated coefficient is insignificant. This low power is also problematic for interpreting the VAR-implied result that excess returns are not an important component of negative returns around inflationary news. This result arises because monthly innovations in the D/P ratio - the variable that drives the unconditional equity risk premium component - are not found to correlate with inflationary surprises. While this is suggestive evidence, due to the low power we do not think we can fully rule out that the equity risk premium is in fact increasing with inflationary news.

One could also conduct a cross section VAR analysis in the spirit of [Vuolteenaho \(2002\)](#), with rolling windows as in [Michael et al. \(2021\)](#) to estimate conditional implications on inflationary news. However, given our caution against over interpreting time series results, we do not pursue this line of analysis in the current paper, instead focusing on survey evidence to validate our decomposition’s cash flow component.

Table A.III: VAR estimation parameters

	(1) Real return	(2) Real rate	(3) Δ bill	(4) Term spread	(5) Div. yield	(6) Rel. bill
L.Real return	0.059 (0.0433)	-0.009*** (0.0021)	0.001*** (0.0003)	0.001 (0.0032)	-0.011*** (0.0008)	0.001*** (0.0003)
L.Real rate	0.369 (0.7090)	0.608*** (0.0344)	-0.011** (0.0047)	-0.001 (0.0530)	-0.001 (0.0132)	-0.010** (0.0042)
L. Δ bill rate	-10.345 (7.2738)	-0.101 (0.3528)	0.256*** (0.0481)	-2.546*** (0.5432)	0.321** (0.1358)	0.265*** (0.0435)
L.Term spread	0.014 (0.2010)	-0.001 (0.0097)	0.002 (0.0013)	0.941*** (0.0150)	0.000 (0.0038)	0.003** (0.0012)
L.Dividend yield	0.252 (0.1814)	0.030*** (0.0088)	0.000 (0.0012)	0.005 (0.0135)	0.994*** (0.0034)	-0.000 (0.0011)
L.Relative bill rate	-2.922 (3.9617)	-0.271 (0.1922)	-0.014 (0.0262)	0.132 (0.2959)	0.176** (0.0740)	0.855*** (0.0237)
Constant	-0.000 (0.0061)	-0.001* (0.0003)	-0.000 (0.0000)	0.001* (0.0005)	0.000 (0.0001)	-0.000 (0.0000)
R^2	0.020	0.443	0.087	0.905	0.994	0.825
N	539	539	539	539	539	539

This table reports the estimated parameter matrix of the first-order VAR [Campbell and Ammer \(1993\)](#) real return decomposition

$$z_t = \mu + A z_{t-1} + w_t$$

where the state vector contains the variables $z_t = [h_t, r_t, \Delta b_t, s_t, dp_t, rb_t]'$ and h_t is the real log market return ($h_t = \ln(1 + R_t^m) - \pi_t$, with π_t the log CPI inflation rate), r_t the real bill rate, Δb_t the change in the nominal bill rate, s_t the term spread (10-year yield minus bill rate), dp_t the dividend-price ratio, and rb_t the relative bill rate (the bill rate less its trailing 12-month moving average). Each column reports the OLS estimates of one equation (the dependent variable named in the column head regressed on the full lagged state z_{t-1}). Standard errors are in parentheses. The model is estimated at the monthly frequency and the sample is 1977 to 2022.

Table A.IV: Return variance decomposition

	Variance	Share
Total, $\text{Var}(\tilde{h})$	0.002469	1.000
Cash-flow news, $\text{Var}(\tilde{\eta}_d)$	0.000903	0.366
Real-rate news, $\text{Var}(\tilde{\eta}_r)$	0.000072	0.029
Excess-return news, $\text{Var}(\tilde{\eta}_e)$	0.000448	0.181
$-2 \text{Cov}(\tilde{\eta}_d, \tilde{\eta}_r)$	0.000233	0.094
$-2 \text{Cov}(\tilde{\eta}_d, \tilde{\eta}_e)$	0.000503	0.204
$+2 \text{Cov}(\tilde{\eta}_r, \tilde{\eta}_e)$	0.000310	0.126

This table reports the *unconditional* variance decomposition of the unexpected real market return implied by the estimated [Campbell and Ammer \(1993\)](#) VAR. The first column reports the variance and covariance terms. The second reports each as a share of $\text{Var}(\tilde{h})$, with the signed terms summing to one. The VAR model is estimated at the monthly frequency and the sample is 1977 to 2022.

Table A.V: Return decomposition around inflationary news using VAR

	Response (pp)	Share
Total real return $\tilde{h}$	-0.9931	1.000
Cash-flow news $\tilde{\eta}_d$	-0.7162	0.721
–Real-rate news $-\tilde{\eta}_r$	-0.4108	0.414
–Excess-return news $-\tilde{\eta}_e$	0.1339	-0.135

This reports the [Campbell and Ammer \(1993\)](#) decomposition of the unexpected real market return into news components, evaluated at the response to a +1 percentage-point CPI surprise in that month. The first column reports news responses in percentage points of the monthly real return. The second column reports each component as a share of the total real return response (columns for the three news terms enter with the signs shown and sum to the total). The VAR model is estimated at the monthly frequency and the sample is 1977 to 2022.

E Movements in the Yield Curve following Inflationary and Monetary News

The unresponsiveness of long-dated real yields to inflation news contrasts with large and positive response of long-dated real yields to monetary policy shocks (Hanson and Stein, 2015). To further explore these distinctive effects of monetary policy news and inflation news on the long end of the real yield curve, we estimate the Hanson and Stein (2015) main regression specification:

$$\Delta f_t^{10,r} = \alpha_r + \beta_r \Delta y_t^2 + \Delta_1 \epsilon_t^r, \quad (22)$$

where $f_t^{10,r}$ is the 10-year $r = \{\text{nominal, real, breakeven inflation}\}$ instantaneous forward rate, y_t^2 is 2-year nominal Treasury yield, and $\Delta x_t = x_{t+1} - x_{t-1}$ is the 2-day change in variable x around an inflation release on the morning of day t . Changes in 2-year nominal yields are used as a measure of monetary policy news that captures surprise changes in both the current federal funds rate and the expected path of the federal funds rate over the next several quarters.

Panel A of Table A.VI presents the results from Equation 22, where columns 1-3 shows the estimation on FOMC days. Consistent with Hanson and Stein (2015), but on an extended sample through to 2022, we find a large impact of monetary policy news on long-dated nominal instantaneous forward rates, with this sensitivity driven by the real rate component of the nominal forward rates.³⁴ Columns 4-6 then shows estimation results on CPI release days. As with FOMC days, there is a large response in long-dated nominal instantaneous forward rates but, in contrast to FOMC days, this sensitivity is driven mostly by the breakeven inflation component of nominal forward rates. For a 100 basis point increase in the 2-year nominal Treasury yield in the 2-days following a CPI release, the nominal instantaneous forward rate increases by 56 basis points, with 40 basis points driven by breakeven inflation, and real forward rates only increasing 17 basis points. The regressor on CPI days can be interpreted as capturing the expected monetary policy response to inflation news on that day, and thus the results point to fundamental difference between news on CPI release days relative to monetary policy days. In particular, the results indicate that yield moves on CPI release days should not be considered just a monetary policy phenomena, i.e. nominal yields increase *only* because of the expected monetary policy response to higher inflation, and instead there are other forces at play driving changes in yields.

To further explore yield curve dynamics in response to various economic shocks, Table A.VI Panel B first estimates Equation 22 on all other days in our sample, i.e. excluding FOMC and CPI release days, and shows that, consistent with Hanson et al. (2021), long-dated nominal instantaneous forward rates are typically highly responsive to moves in short-dated nominal rates, with the majority of the sensitivity driven by the real rate component of the nominal rate, but also a role for breakeven inflation component. More importantly, Columns 4-9 of Panel B Table A.VI next shows the results splitting other days into monetary policy news days and growth news days.³⁵

³⁴Using different shocks for monetary policy, Nakamura and Steinsson (2018) do not find an impact of monetary policy shocks on 10-year real forward rates, but do find monetary policy effects 5-year forward rates and other shorter maturities of the real yield curve.

³⁵We split days into two groups conditional on the correlation of yields and stock returns on that day: days when stock returns and yields are negatively correlated are labeled monetary policy news days, and days when stock returns and yields are positively correlated are labeled growth dates. This split follows a recent literature (Cieslak and Schrimpf, 2019; Jarociński and Karadi, 2020; Cieslak and Pang, 2021; Hoek

Strikingly, yield curve dynamics on monetary policy (but non-FOMC) days exhibit very similar behavior as on FOMC days themselves, with nominal forward rates purely driven by real rates. On growth news days, nominal forward rates are mostly driven by real rates, but there is a role for breakeven inflation too. Nevertheless, CPI releases standout, even relative to growth days, as days when the long-end of the nominal yield curve is particularly driven by inflation compensation changes. The results therefore indicate that particular economic channels are at play on CPI days and are consistent, for example, with a model in which long-run inflation expectations are not well anchored and revise in light of incoming inflationary news ([Gürkaynak, Sack and Swanson, 2005](#)).

et al., [2022](#)) that uses the intuition that days with a positive correlation between yields and stock returns must contain positive growth news. Increasing yields increase the discounting of expected cash flows and thus, for stock returns to be positive, there must also be positive news about expected cash flows.

Table A.VI: Forward yields around monetary and inflationary news

Panel A: FOMC and CPI release days									
	FOMC			CPI releases					
	Nominal	Real	Inflation	Nominal	Real	Inflation			
2-year treasury	0.49*** (0.15)	0.50*** (0.12)	-0.01 (0.09)	0.56*** (0.10)	0.17** (0.08)	0.40*** (0.09)			
R-squared	0.068	0.104	0.000	0.131	0.019	0.090			
N	146	146	146	218	218	218			

Panel B: All other days									
	All other days			Monetary news days			Growth news days		
	Nominal	Real	Inflation	Nominal	Real	Inflation	Nominal	Real	Inflation
2-year treasury	0.58*** (0.02)	0.45*** (0.02)	0.14*** (0.02)	0.46*** (0.04)	0.46*** (0.03)	0.01 (0.03)	0.63*** (0.02)	0.44*** (0.02)	0.19*** (0.02)
R-squared	0.159	0.118	0.018	0.085	0.099	0.000	0.202	0.129	0.041
N	4,226	4,226	4,226	1,662	1,662	1,662	2,564	2,564	2,564

This table shows results from Equation 22:

$$\Delta_1 f_t^{10,r} = \alpha_r + \beta_r \Delta_1 y_t^2 + \epsilon_t^r,$$

where $f_t^{10,r}$ is the 10-year $r = \{\text{nominal, real, breakeven inflation}\}$ instantaneous forward rate, y_t^2 is 2-year nominal Treasury yield, and $\Delta_1 x_t = x_{t+1} - x_{t-1}$ is the 2-day change in variable x_t . The regressions are estimated over the sample period 2004–2022 and robust standard errors are in parentheses. Panel A shows estimation results on FOMC days and CPI release days separately. Panel B shows results on all other non-FOMC and non-CPI release days, before splitting all days into 'monetary news' days and 'growth news' days, where days are assigned conditional on the correlation between stock returns and nominal yields. *** p<0.01, ** p<0.05, * p<0.1.

F Time Series and State Dependence

In the baseline analysis, we showed the coefficients of interest varied during the GFC and in periods of elevated oil volatility. This raises the question of how sensitive our main findings are to specific periods or states. This section therefore expands on the state-dependence analysis presented in [Table V](#).

Time series dependence. We first re-estimate [Equation 12](#) using 5-year rolling estimation windows. We run the rolling regressions with the real return on the aggregate stock market as the dependent variable, before cycling through each of the real yield curve, the real cash flow, and the equity premium return component as the dependent variable. The exercise is thus similar to that in [Figure II](#), but based on rolling regression estimates.

[Figure A1](#) presents the rolling decomposition results for 1-day real returns in response to inflationary news. It shows that in 97 percent of the 222 estimation windows, the stock market is estimated to have declined with inflationary news, highlighting that the fact that higher-than-expected (lower-than-expected) inflation leads to negative (positive) stock returns is a highly robust feature of the data. That said, the coefficient is not consistent over time, with the stock market being particularly sensitive to inflationary news in the 5-year sample windows ending in 2004-2006, 2013-2016 and 2021-2022. The estimated coefficient is less than -2 percent in each of these sample windows.

[Figure A1](#) also documents interesting variation in the component drivers of the stock market's negative response to inflationary news. The real cashflow component is negative in 89 percent of the estimation windows, and thus is nearly always a negative contributor, while the real yield curve is only a negative contributor in 15 percent of the window periods. This negative contribution of the real yield curve occurred at the end of our sample, as the expected monetary response to inflationary news in the above target inflationary regime of the post-COVID-era likely led investors to increase their expectations of future real yields. For the equity premium component of the stock return, the time series variation in the estimated coefficient is much more pronounced, with a negative coefficient estimated in 57 percent of the sample windows. For the windows ending in 2015-2016, equity premium particularly weighed on the stock market response to inflationary news, with the most negative equity premium return component estimates in these periods.

State dependence. To test for further state dependence beyond what was shown in [Table V](#), we adjust [Equation 12](#) with an additional interaction term between inflationary news and the series of other variables that measure economically interesting states of the economy:

$$Return_t = \alpha + \beta_1 \text{Inflationary News}_t + \beta_2 \text{State}_t + \beta_3 \text{Inflationary News}_t \times \text{State}_t + \epsilon_t. \quad (23)$$

Our coefficient of interest is β_3 , which captures the marginal stock market effect of being in a particular state of the economy with respect to an inflationary shock. We use several variables to define the state of the economy, as described below, and present results from [Equation 23](#) across all measures of $State_t$ in [Table A.VII](#). In summary, we do not find evidence of state dependence.

Our first set of conditioning variables therefore focuses on whether the current economic situation is dominated by demand- or supply-driven shocks. First, we compute the correlation between

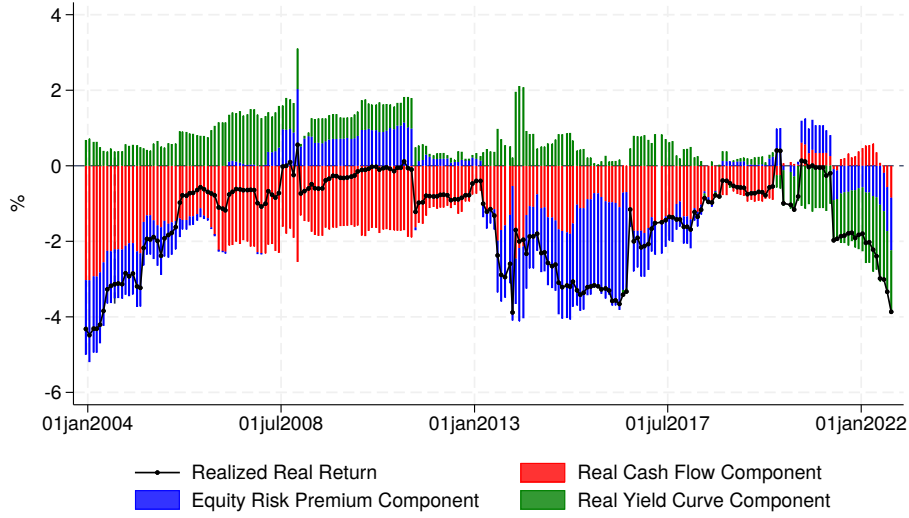
daily stock and bond returns using an exponentially weighted moving average with 75% of the weight distributed over the most recent 22 days. Second, we use the inflation risk premium derived from the no-arbitrage term structure model of [d’Amico et al. \(2018\)](#) to measure the premium investors require for the possibility that inflation may rise or fall more than they expect over the period in which they hold a bond. Third, we use a measure that separates inflation into its supply and demand factors from the San Francisco Federal Reserve Bank. Proposed by [Shapiro \(2022\)](#), this measure seeks to separate the impact of supply and demand factors on monthly inflation. To do this, inflation rates are classified by spending category, and these categories are then divided into supply and demand-driven groups. Demand-driven categories take place when surprising price changes happen in the same direction as the unexpected change in quantity. Supply-driven categories take place when surprising price changes move in the opposite direction as the quantity changes.

We now shift focus away from whether the economy is demand or supply driven and consider other ways in which the economy could be state dependent. First, the effects of inflation on the economy may not be linear ([Fischer, 1993](#)). When inflation is very low, higher inflation may be associated with better economic outcomes, e.g. in a secular stagnation environment, but when inflation exceeds a certain threshold the cost of inflation can become very large, as price dispersion increases rapidly with inflation. In column (5) we test for the non-linear effects of inflationary news on the stock market by interacting the inflation shock with the 3-month moving average inflation print. Second, in column (6) we test for the asymmetric effects of inflation. While more recently inflation has surprised to the upside, in the post-GFC period inflation has often surprised to the downside. To test for whether higher-than-expected inflation leads to stock market losses or lower-than-expected inflation leads to stock market gains, we define a dummy that is 1 if inflationary news is positive. Third, in column (7) we test whether variation in the state of the business cycle, measured by the unemployment gap, can explain time-varying effects of inflation on the stock market. Finally, we test whether perceptions about monetary policy responsiveness has been driving the sensitivity of the stock market to inflation news, as suggested in [Rubio Cruz et al. \(2023\)](#) for the most recent period, and evidence shown in ([Boyd et al., 2005](#); [Elenov et al., 2024](#)) for other macro news. To test this hypothesis, our variable takes the value of the change in the 2 year nominal and real bond yield in columns (8) and (9), respectively.

Across specifications, we find negative stock returns in response to inflationary news is a generally robust result. Outside of the GFC period, the response has been uniformly negative.³⁶ This holds even with conditioning on state variables that attempt to measure the degree to which the economy was demand-driven, further supporting a flat Phillips curve interpretation that only with very large demand shocks will inflation become demand-driven.

³⁶In unreported results, we also show the results are not driven by the zero lower bound (ZLB) period. In fact, consistent with [Gourio and Ngo \(2020\)](#), we find full sample estimates are somewhat attenuated by including this period, and therefore results are even stronger with the ZLB excluded.

Figure A1: Rolling decomposition of real stock returns around inflationary news



This figure stacks the rolling time series of the estimated beta coefficient from Equation 12:

$$y_t^c = \alpha^c + \beta^c \text{Inflationary News}_t + \epsilon_t^c,$$

where y_t^c is the predicted $c = \{\widetilde{CF}, \widetilde{ERP}, \widetilde{YC}\}$ component of the 1-day real return on the day of the CPI announcement on date t , and the rolling estimation window is 5-year periods. The return components are the returns generated from changes in real cash flow expectations, $\widetilde{CF}$, changes in equity risk premium, $\widetilde{ERP}$, and changes in real risk-free rates, $\widetilde{YC}$. The black line in the figure plots the rolling coefficients of 1-day real realized returns regressed on inflationary news, which is by definition the sum of the stacked coefficients on the return components. Inflationary News $_t$ is the difference between the published month-on-month CPI inflation and the median expectation of forecasts ahead of the announcement. The overall sample period is 1999–2022, which means the first 5-year window period in 2004.

Table A.VII: State Dependence

	Dependent Variable: 1-day $Return_{i,t}$								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Inflationary News	-0.928*** (0.307)	-0.983*** (0.333)	-1.012*** (0.379)	-1.368** (0.590)	-0.913*** (0.323)	-0.614 (0.576)	-0.896*** (0.310)	-1.371** (0.610)	-0.909*** (0.311)
State		0.0309 (0.0442)	0.0815* (0.0488)	-0.0413 (0.0763)	-0.0340 (0.0431)	0.180 (0.165)	-0.0333 (0.0428)	-0.0502 (0.0856)	-0.0139 (0.0438)
Inflationary News $\times$ State		0.0994 (0.245)	0.809** (0.400)	0.876 (0.715)	0.00310 (0.247)	-1.439 (1.008)	0.348 (0.339)	-0.260 (0.746)	-0.165 (0.282)
R-squared	0.017	0.018	0.029	0.024	0.018	0.021	0.017	0.018	0.018
N	529	504	467	279	528	529	528	279	529
State		Stock-Bond	Inflation RP	Supply	Inflation	Positive	U-gap	$\Delta_0 2yr$	$\Delta_0 2yn$

This table shows the coefficient estimates from regression [Equation 23](#):

$$Return_{i,t} = \alpha + \beta_1 \text{Inflationary News}_t + \beta_2 \text{State}_t + \beta_3 \text{Inflationary News}_t \times \text{State}_t + \epsilon_{i,t},$$

where $Return_{i,t}$ is the average stock return between the close of business the day before the CPI announcement on date t and close of business on the day of release for stock i . $\text{Inflationary News}_t$ is the difference between the published month-on-month CPI inflation and the median expectation of forecasts ahead of the announcement. β_3 captures the marginal stock market effect of being in a particular state of the economy with respect to an inflationary shock. Robust standard errors in parenthesis. Stock-Bond is the standardized correlation between stock and bond returns. Inflation RP is a measure of the inflation risk premium from [d'Amico et al. \(2018\)](#). Supply is the standardized share of inflation that is supply-driven from [Shapiro \(2022\)](#). Inflation is the last inflation print. Positive is a dummy if the inflation surprise is positive. U-gap is the unemployment gap. $\Delta_0 2yr$ and $\Delta_0 2yn$ are the same day change in the 2 year real and nominal bond yield, respectively. Robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

G A Simple Model of a Firm Facing a Cost Shock

In standard industrial organization models (Tirole, 1989), firm profit is

$$\Pi(p) = (p - c)Q(p) \quad (24)$$

where p is the product price, c is the marginal cost of production, and $Q(p)$ is the quantity produced. First order conditions with respect to the firm's pricing decision imply that the equilibrium price is

$$p = \frac{\epsilon}{\epsilon - 1}c, \quad (25)$$

where $\epsilon = -\frac{\partial \ln Q(p)}{\partial \ln p} > 1$ is the elasticity of demand facing the firm.

Defining the markup $m = \frac{\epsilon}{\epsilon - 1} > 1$ and substituting the equilibrium price Equation 25 into Equation 24, we find that firm profit in equilibrium is thus

$$\Pi^* = (m - 1)cq, \quad (26)$$

where $q = Q(p^*)$ is the equilibrium quantity produced.

We now consider the impact of a change in marginal costs on firm profits. By the envelope theorem, the effect of a change in marginal cost on equilibrium profits is

$$\begin{aligned} \frac{\partial \Pi^*}{\partial c} &= \left. \frac{\partial \Pi(p, c)}{\partial c} \right|_{p=p^*} \\ &= -q, \end{aligned} \quad (27)$$

where $q = Q(p^*)$ is equilibrium quantity. Intuitively, holding the firm's optimal pricing response into account, a marginal increase in the cost of producing each unit reduces profits by the equilibrium quantity sold.

Multiplying both sides by $\frac{c}{\Pi^*}$ gives the proportional response of profits to marginal costs:

$$\begin{aligned} \frac{\partial \Pi^* / \Pi^*}{\partial c / c} &= -\frac{cq}{(p - c)q} \\ &= -\frac{c}{p - c} \\ &= -\frac{1}{m - 1} \\ &= 1 - \epsilon < 0, \end{aligned} \quad (27)$$

where the third line uses $p = mc$ and the fourth uses $m = \frac{\epsilon}{\epsilon - 1}$.

Equation (27) gives the first key testable prediction: given $\epsilon > 1$, firm profits decline following an increase in marginal costs.

The second key testable prediction follows by differentiating the proportional sensitivity of profits to marginal costs with respect to the elasticity of demand:

$$\frac{\partial}{\partial \epsilon} \left(\frac{\partial \Pi^* / \Pi^*}{\partial c / c} \right) = -1. \quad (28)$$

We therefore see that firms with higher elasticity of demand—i.e., less market power—suffer larger proportional declines in profitability in response to marginal cost shocks than firms with lower elasticity of demand.

Table A.VIII: Summary statistics

Panel A: Cross section variables								
	N	Mean	SD	p10	p25	p50	p75	p90
1-day stock return (perc.)	2,320,391	0.05	4.69	-3.57	-1.37	0.00	1.28	3.57
5-day stock return (perc.)	2,320,248	0.36	9.61	-8.12	-3.33	0.05	3.54	8.68
Markup	2,320,527	1.73	1.74	0.93	1.07	1.29	1.76	2.75
Size: ln(assets)	2,320,422	5.51	2.37	2.47	3.77	5.42	7.15	8.62
Book-to-market ratio	2,288,857	0.69	2.97	0.14	0.30	0.56	0.93	1.45
Leverage	2,310,867	0.23	0.24	0.00	0.04	0.18	0.35	0.51
Asset tangibility	2,303,175	0.26	0.24	0.02	0.06	0.18	0.39	0.66
Stock market beta	1,673,444	0.99	0.71	0.24	0.57	0.94	1.34	1.80
SMB risk factor beta	1,673,444	0.80	1.14	-0.25	0.15	0.66	1.29	2.05
HML risk factor beta	1,673,444	0.15	1.13	-1.06	-0.38	0.20	0.72	1.27

Panel B: Time-series variables (percentages)								
	N	Mean	SD	p10	p25	p50	p75	p90
Inflationary news	536	-0.00	0.14	-0.20	-0.10	0.00	0.10	0.20
1-day return (CRSP equal-weighted portfolio)	536	0.05	1.14	-1.10	-0.44	0.11	0.57	1.15
1-year equity risk premium (lower bound)	438	4.03	2.06	2.14	2.66	3.58	4.87	6.17
2-year nominal Treasury yield	536	4.91	3.72	0.40	1.51	4.72	7.47	10.04
10-year nominal Treasury yield	536	5.98	3.22	2.03	3.09	5.58	8.25	10.70
2-year real Treasury yield	283	0.34	1.68	-1.49	-0.83	0.08	1.26	3.03
10-year real Treasury yield	283	1.36	1.40	-0.55	0.40	1.14	2.30	3.46
2-year Treasury breakeven inflation	283	1.78	0.97	0.92	1.41	1.79	2.35	2.76
10-year Treasury breakeven inflation	283	2.13	0.42	1.57	1.85	2.20	2.44	2.60

This table presents summary statistics of the variables used in the empirical analysis. Panel A shows panel variables that are available in the cross section of firms, and Panel B shows time-series variables. HML is high minus low. SMB is small minus big. The full sample extends from 1977 through 2022 with 12 observations per calendar year that correspond to the monthly CPI releases, as published by the Bureau of Labor Statistics. Inflationary news are measured as the difference between the published month-on-month CPI inflation and the median expectation of forecasts ahead of the announcement. For full information on data sources and variable construction, refer to [section 3](#).

Table A.IX: Stock returns, inflation expectations and inflationary news

	(1) Inflation Expectations	(2) Return	(3) Return	(4) Return
Inflation News	0.074*** (0.025)	-0.014* (0.008)		
Inflation Expectations			0.070** (0.029)	
Inflation Expectations (Predicted)				-0.193 (0.150)
Constant	-0.000 (0.003)	-0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)
R-squared	0.041	0.017	0.058	.
First-stage F-statistic				8.653
N	285	285	285	285

Inflationary News is the difference between the published month-on-month CPI inflation and the median expectation of forecasts ahead of the announcement. Inflation Expectations are the 1-day change in 10-year inflation expectations. Return is the 1-day S&P 500 return. Inflation Expectations (Predicted) are the change in inflation expectations instrumented with inflationary news. Robust standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table A.X: Core inflation surprises

			2-year maturity			10-year maturity		
	Return	Δ ERP	Nominal	Inflation	Real	Nominal	Inflation	Real
Core News	-1.24** (0.49)	0.27* (0.14)	0.15*** (0.03)	0.30*** (0.10)	-0.20** (0.10)	0.12*** (0.03)	0.13*** (0.02)	-0.03 (0.03)
R-squared	0.016	0.010	0.079	0.033	0.014	0.041	0.087	0.004
N	387	387	387	278	278	387	278	278

Core News is the difference between the published month-on-month *core* CPI inflation and the median expectation of forecasts ahead of the announcement. Return is the S&P 500 return, Δ ERP is the change in the equity risk premium. *Nominal* is the change in nominal yields. *Inflation* is the change in inflation expectations. *Real* is the change real yields. Robust standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table A.XI: Drivers of daily equity premium changes around inflationary news

	(1) risk aversion	(2) economic uncertainty	(3) monetary policy uncertainty
Inflationary News	7.46 (5.18)	2.97* (1.77)	2.81*** (0.88)
Constant	-0.01 (0.01)	-0.00 (0.00)	-0.00 (0.00)
R-squared	0.012	0.014	0.039
N	283	283	283

This table shows estimation results from the following equation:

$$\Delta y_t = \alpha + \beta \text{Inflationary News}_t + \epsilon_t,$$

where $\Delta y_t = y_t - y_{t-1}$ is the 1-day change in variable y around Inflationary News_{*t*} that is released at 8:30am on day t . The variables considered are the [Bekaert et al. \(2022\)](#) measures of risk aversion and economic uncertainty, and the Kansas City Fed's Measure of Policy Rate Uncertainty (KC PRU). Robust standard errors are reported in parentheses. The sample period is 1999-2022. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A.XII: The yield curve and inflationary News—evidence from swap rates

	2-year maturity			10-year maturity		
	Nominal	Inflation	Real	Nominal	Inflation	Real
Inflationary News	0.13*** (0.04)	0.27*** (0.05)	-0.14** (0.07)	0.10** (0.04)	0.09*** (0.03)	0.01 (0.05)
R-squared	0.071	0.176	0.040	0.039	0.074	0.000
N	220	220	220	220	220	220

This table shows coefficient estimates from regression [Equation 12](#):

$$\Delta y_t = \alpha + \beta \text{Inflationary News}_t + \epsilon_t,$$

where $\Delta y_t = y_t - y_{t-1}$ is the 1-day change in the 2-year or 10-year yield from the day before the CPI announcement on date t . Inflationary News_{*t*} is the difference between the published month-on-month CPI inflation and the median expectation of forecasts ahead of the announcement. The table shows results for 2-year and 10-year yields on interest rate swaps (nominal), inflation swap rates (inflation), and the swap-implied real yield (the interest rate swap yield minus the inflation swap yield). Swap data is taken from Bloomberg, and the sample period extends from 2004 through 2022. Robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A.XIII: Perceived comovement of revisions in SPF inflation and growth expectations, and its state dependence

	Baseline	GFC	GFC + oil vol.
Δ Expected growth	0.000 (0.014)	-0.007 (0.014)	0.024 (0.015)
Δ Expected growth $\times$ GFC		0.214*** (0.065)	0.193*** (0.068)
Δ Expected growth $\times$ oil volatility			-0.014*** (0.003)
Observations	4,825	4,825	4,478
R^2	0.123	0.127	0.121

Notes. The table reports the perceived Phillips slope of professional forecasters, estimated from individual-level Survey of Professional Forecasters (SPF) data. The dependent variable is the quarter-to-quarter revision in a forecaster's four-quarter-ahead expected CPI inflation. Column 1 reports the unconditional slope. Column 2 adds an interaction with a Global Financial Crisis indicator (2007q4–2009q2). Column 3 adds, jointly, interactions with the GFC indicator and with oil-price volatility, measured as the one-quarter-lagged, standardized quarterly volatility of daily oil-futures returns. All columns include forecaster and quarter fixed effects, and standard errors (in parentheses) are two-way clustered on forecaster and quarter. Stars denote significance at the 10%, 5%, and 1% levels (*, **, ***).

Table A.XIV: Inflationary news, cost elasticity, and stock returns

	Dependent Variable: $Return_{i,t}^5$					
	(1)	(2)	(3)	(4)	(5)	(6)
Inflationary News	-1.138 (1.249)					
Cost Elasticity	0.00560*** (0.000317)	0.00650*** (0.000294)	0.00675*** (0.000473)	0.00721*** (0.000743)	0.00876*** (0.000907)	0.00916*** (0.00127)
Inflationary News $\times$ Cost Elasticity	0.0665*** (0.00348)	0.0712*** (0.000491)	0.0671*** (0.00105)	0.0666*** (0.00335)	0.0699*** (0.00273)	0.0698*** (0.00473)
R-squared	0.014	0.142	0.176	0.176	0.185	0.185
N	935,259	935,259	933,018	918,065	765,095	754,113
Firm FE	✓	✓	✓	✓	✓	✓
Time FE		✓	-	-	-	-
Industry-Time FE			✓	✓	✓	✓
Int. Firm Controls				✓		✓
Int. Factor Controls					✓	✓

This table shows results from [Equation 13](#):

$$Return_{i,t}^5 = \alpha + \beta_1 \text{Inflationary News}_t * \text{Cost Elasticity}_{i,y(t)-1} + \alpha_i + \alpha_t + \mathbf{X}'\gamma + \epsilon_{i,t},$$

where $Return_{i,t}^5$ is the cumulative stock returns calculated from day t-1, before CPI release, to day t+5, after the CPI release. $\text{Inflationary News}_t$ is the difference between the published month-on-month CPI inflation and the median expectation of forecasts ahead of the announcement. $\text{Cost Elasticity}_{i,y(t)-1}$ is the percent change in economic profits of the firm following a 1% rise in all firms' marginal costs and standardized to have mean 0 and a standard deviation of 1. α_i is a firm fixed effect. α_t is a date fixed effect. $\mathbf{X}$ are controls. *Int. Firm Controls* includes firm characteristics controls—log assets, tangibility, leverage, and market-to-book value—interacted with Inflationary News. *Int. Factor Controls* includes firm-level rolling-betas to the [Fama and French \(1993\)](#) asset pricing factors—market beta, size, and value—each interacted with Inflationary News. Standard errors (in parentheses) are double clustered at the firm and date level. *** p<0.01, ** p<0.05, * p<0.1.

Table A.XV: Propensity Score Matching: Inflation $\times$ Markup Interaction

	(1) Full sample	(2) Decile	(3) Quartile	(4) Tercile	(5) Median
Inflation Surprise $\times$ Markup	0.226* (0.130)	0.305** (0.127)	0.245** (0.122)	0.251* (0.128)	0.219* (0.131)
Observations	2,208,380	333,266	837,809	1,121,201	1,694,717
R^2	0.128	0.178	0.149	0.139	0.124
<i>Covariate balance (mean standardized % bias)</i>					
Before matching		11.8	15.2	14.2	10.8
After matching		1.7	2.4	1.8	1.2

Notes: The dependent variable is the five-day cumulative stock return around the inflation (CPI) announcement. Column (1) is the baseline specification on the full sample. Columns (2)–(5) re-estimate the interaction on samples in which high-markup firms are matched to low-markup firms on observable characteristics. Treatment/control groups are the top vs. bottom decile, quartile, tercile, and half of the markup distribution. Matching is one-to-one nearest-neighbor on the estimated propensity score (logit) with common support and a caliper of 0.01; matching covariates are firm size (log assets), book-to-market, leverage, tangibility, and CAPM, SMB, and HML factor loadings. All specifications include firm and date fixed effects; standard errors (in parentheses) are two-way clustered by firm and date. The balance panel reports the mean absolute standardized percentage bias across covariates before and after matching. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.XVI: Falsification Tests: Markup Interaction Across Macro Surprises

	(1) Baseline CPI inflation	(2) Falsification shocks Employment (NFP)	(3) Falsification shocks Monetary policy (FOMC)
Surprise $\times$ Markup	0.038** (0.016)	-0.006 (0.015)	-0.028 (0.027)
Observations	2,283,661	1,728,682	1,272,662
R^2	0.128	0.173	0.140

Notes: Each column interacts firm markup with a different macro surprise, using an identically constructed five-day cumulative return around the respective announcement as the dependent variable and the same standardized, firm-level measure of markup. Column (1) is the baseline headline CPI inflation surprise. Columns (2) and (3) are falsification shocks: the nonfarm-payroll (employment) surprise and the high-frequency monetary-policy surprise of Bauer and Swanson around FOMC announcements. All specifications include firm and date fixed effects; standard errors (in parentheses) are two-way clustered by firm and date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.XVII: Frequency of price adjustments, stock returns and inflationary news

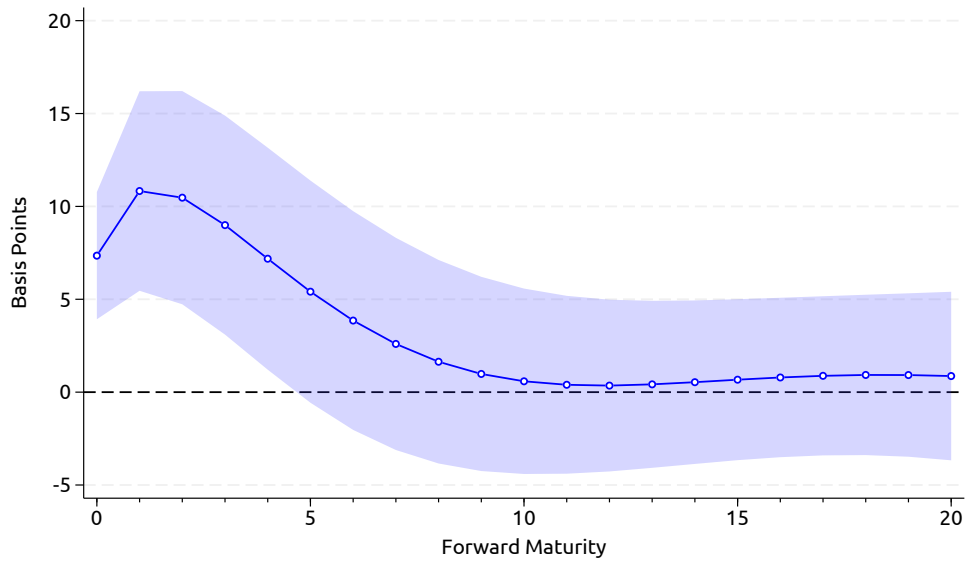
	Dependent Variable: $Return_{i,t}^5$		
	(1)	(2)	(3)
FPA	-0.0918 (0.182)		-0.102 (0.180)
Inflationary News $\times$ FPA	1.398 (1.118)		1.276 (1.124)
Markup		-0.000807 (0.00925)	-0.000612 (0.00922)
Inflationary News $\times$ Markup		0.165** (0.0805)	0.177** (0.0793)
R-squared	0.152	0.155	0.154
N	2,317,305	2,283,628	2,150,311
Firm FE	✓	✓	✓
Time FE	-	-	-

This table shows results from the following equation:

$$Return_{i,t}^5 = \alpha + \beta_1 \text{Inflationary News}_t * \text{Markup}_{i,y(t)-1} + \beta_2 \text{Inflationary News}_t * \text{FPA}_i + \alpha_i + \alpha_t + \mathbf{X}'\gamma + \epsilon_{i,t},$$

where $Return_{i,t}^5$ is the cumulative stock returns calculated from day $t-1$, before CPI release, to day $t+5$, after the CPI release. $\text{Inflationary News}_t$ is the difference between the published month-on-month CPI inflation and the median expectation of forecasts ahead of the announcement. $\text{Markup}_{i,y(t)-1}$ is the estimated markup from [De Loecker et al. \(2020\)](#) and is standardized to have mean 0 and a standard deviation of 1. FPA_i is the sectoral frequency of price adjustment from [Pasten et al. \(2020\)](#), merged to firms at the six-digit NAICS level where available and otherwise at the four- or two-digit level; higher values indicate more flexible (less sticky) prices, so a positive interaction with Inflationary News means flexible-price firms decline by less. α_i is a firm fixed effect. α_t is a date fixed effect. $\mathbf{X}$ are controls. *Int. Firm Controls* includes firm characteristics controls—log assets, tangibility, leverage, and market-to-book value—interacted with Inflationary News. *Int. Factor Controls* includes firm-level rolling-betas to the [Fama and French \(1993\)](#) asset pricing factors—market beta, size, and value, each interacted with Inflationary News. The sample period is 1977–2022. Standard errors (in parentheses) are double clustered at the firm and date level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure A2: Forward nominal yields around inflationary news

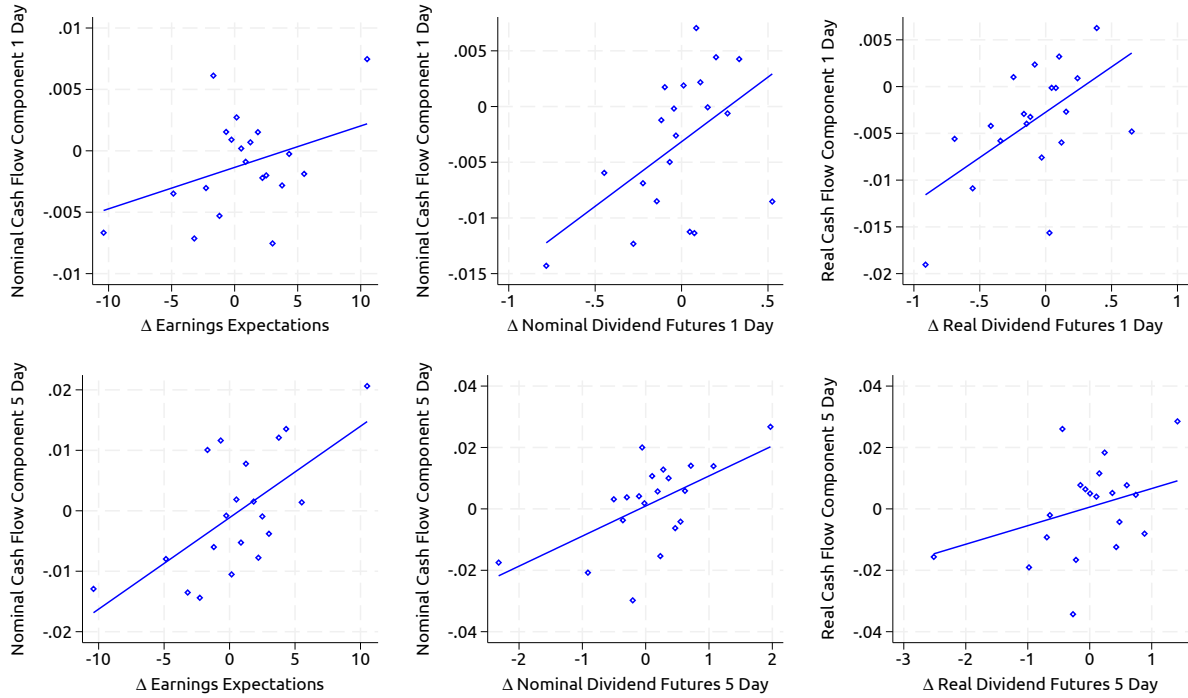


This figure plots the estimated coefficient of the regression specification:

$$\Delta f_t^{(j)} = \alpha^j + \beta^j \text{Inflation News}_t + \epsilon_t^j,$$

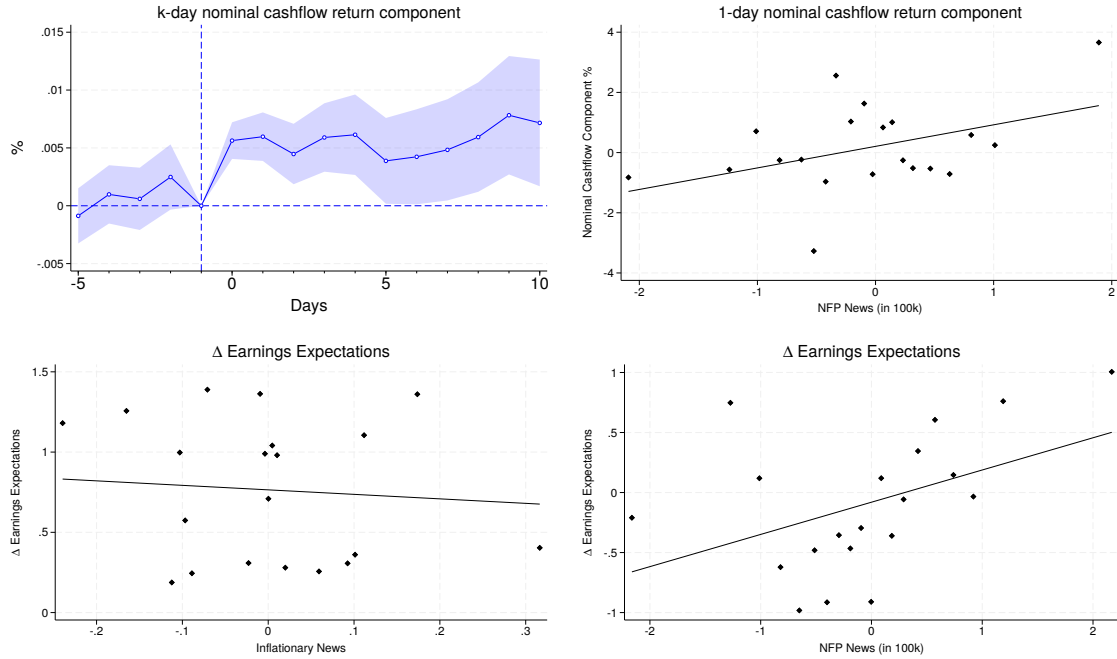
where $\Delta f_t^{(j)} = f_t^{(j)} - f_{t-1}^{(j)}$ is the 1-day change in the j -year forward 1-year nominal yield. The x-axis label denotes the number of years ahead the 1-year forward rate refers to. For $j = 0$, we use the raw 1-year nominal yield, while for all other $j > 0$ we use the 1-year yield in j years' time. The sample period is 1999 to 2022. The shaded area reflects the 90% confidence interval.

Figure A3: Cash flow component validation: correlation with direct cash flow measures



This figure combines binscatterplots between the 1- (5-) day extracted cash flow component of stock returns, as defined in [section 4](#), on the y-axis on the upper (lower) panels against alternative measures of cash flow expectations. In the left panels we plot nominal extracted earnings expectations against $\Delta Earnings Expectation_t$ and $\Delta_k \ln(Div.Futures_t^2)$ (middle panel) on the x-axis. The right panels plot the real cash flow component of stock returns and real dividend futures on the x-axis. $\Delta Earnings Expectation_t$ is the (asset-) weighted average change in earnings expectations around CPI dates, as described in [Appendix A](#). $\Delta_k \ln(Div.Futures_t^2)$ are risk premium-adjusted log changes in the 2-year-ahead dividend future price over period k , as described in [subsection 3.2](#).

Figure A4: Cash flow component validation: cash flow expectations around economic activity news

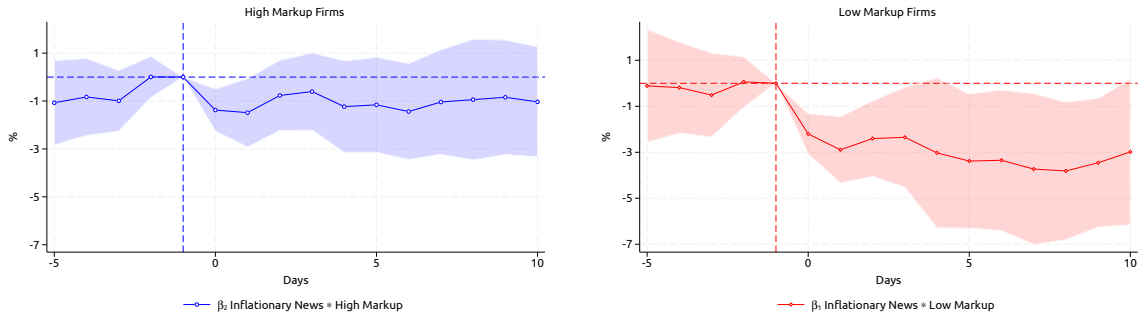


The top left figure plots the estimated coefficient of the equation:

$$y_t^k = \alpha^k + \beta^k \text{NFP News}_t + \epsilon_t^k,$$

where y_t^k is the nominal cash flow return component from the stock return decomposition from the day before the nonfarm payroll announcement on date t to k days after the announcement. NFP News_t is the difference between the published monthly rise in total nonfarm payroll employment and the median expectation of forecasts ahead of the announcement (in 100k). The shaded area reflects the 90% confidence interval. The top right figure presents a binscatterplot between the 1-day nominal cash flow return component, y_t^1 , on the y-axis against the NFP News_t on the x-axis. The sample period of the top row is 1999–2022. The bottom row presents binscatterplots between $\Delta \text{Earnings Expectation}_{i,t}$ on the y-axis and Inflationary News $_t$ (left figure) and NFP News_t (right figure) on the x-axis, respectively. $\Delta \text{Earnings Expectation}_{i,t}$ is the log change in earnings expectations between 15 days after and before the economic news release. The black line plots the linear fit across all releases.

Figure A5: Real estimated cash flows around inflationary news by market power



This figure plots the estimated coefficient of Equation 15:

$$\begin{aligned} \widehat{Return}_{i,t}^{k,\widehat{CF}} = & \alpha + \beta_1 \text{Inflationary News}_t * \text{Low Markup}_{i,y(t)-1} \\ & + \beta_2 \text{Inflationary News}_t * \text{High Markup}_{i,y(t)-1} + \beta_3 \text{Low Markup}_{i,y(t)-1} + \epsilon_{i,t}, \end{aligned}$$

where $\widehat{Return}_{i,t}^{k,\widehat{CF}}$ is the estimated real cash flow component of stock returns between the day before the CPI announcement on date t and k days after for stock i . Inflationary News $_t$ is the difference between the published month-on-month CPI inflation and the median expectation of forecasts ahead of the announcement.

High Markup $_{i,y(t)-1}$ is a dummy that is equal to 1 if the firm has a markup above the 75th percentile of the distribution and 0 otherwise. Low Markup $_{i,y(t)-1}$ is a dummy that is equal to 1 if the firm has a markup below the 25th percentile of the distribution and 0 otherwise. Markup is defined as the estimated markup from De Loecker et al. (2020). Standard errors are double clustered at the firm and date level. The shaded area reflects the 90% confidence interval. The sample period is 1999–2022.